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Numerical investigation of the transport mechanisms governing arc - keyhole interaction in deep-penetration K-TIG welding under the influence of metal vapour

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Abstract

In keyhole tungsten inert gas (K-TIG) welding, the use of high welding currents inevitably causes partial vaporization of the base metal (BM), generating metal vapour that markedly influences arc plasma transport phenomena. In the present work, the BM surface was assumed to remain undeformed, and keyholes of varying depths were predefined to accelerate the numerical computation. Under this assumption, the effects of metal vapour on arc plasma characteristics – including temperature distribution, current density, velocity, pressure field, electromagnetic force, and plasma flow – were investigated at different keyhole depths. The presence of metal vapour was found to increase electrical conductivity within the medium-temperature range (5000–15,000K), thereby altering current density distribution and reducing both Joule heating and

Outline

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CRediT authorship contribution statement

Consent to participate

radiation losses. As a result, the maximum arc plasma temperature decreased from approximately 19,500K to about 14,500K. Consequently, the low - temperature isotherms near the tungsten electrode tip expanded outward, whereas the high - temperature core region and the isotherms adjacent to the BM contracted inward. With increasing keyhole depth, the arc plasma penetrated further into the keyhole, increasing its length and thus raising the potential difference. However, when metal vapour was considered, the changes in electrical conductivity caused the potential difference to be about 2.0V lower than that without vapour. The current was predominantly concentrated near the entrance of the keyhole wall, where intense Joule heating promoted BM vaporization and further metal vapour accumulation. Electrical conductivity various cause redistribution of current density reduced both the axial and radial components of the electromagnetic force. Simultaneously, the arc plasma pressure decreased from approximately 1700Pa to about 1400Pa. The combined reduction in electromagnetic force and arc plasma pressure lowered the axial velocity of the arc plasma by roughly $200\text{m}\cdot\text{s}^{-1}$ after keyhole formation. Moreover, metal vapour diminished plasma drag shear stress and modified the Marangoni shear stress distribution, thereby affecting weld pool flow patterns and keyhole morphology. To evaluate the validity of the undeformed - BM assumption with preset keyholes, a fully coupled arc plasma - keyhole - weld pool model was established to capture dynamic keyhole evolution. The simulated keyhole profiles showed strong agreement with the preset geometries. Furthermore, comparisons between simulated and experimentally captured arc plasma profiles at representative keyhole depths (0mm and 10mm) exhibited high consistency, confirming the accuracy and reliability of the proposed simulation approach.

Keywords

K-TIG welding; Arc physics; Metal vapour; Numerical simulation

[Consent for publication](#)

[Ethics approval](#)

[Funding](#)

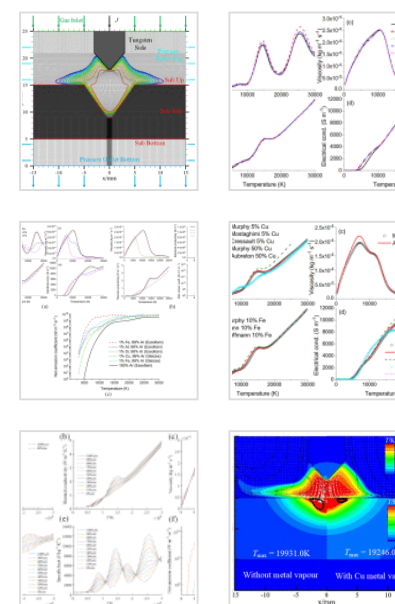
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1. Introduction

K-TIG welding has emerged as a highly efficient and cost-effective process in modern welding technology, particularly for thick plate welding [1]. Compared to other welding methods such as GMAW and GTAW, K-TIG welding provides superior mechanical properties in welded joints [2,3]. Over the past few years, considerable attention has been devoted to optimizing K-TIG welding parameters and expanding its use in various material systems, including low-carbon steel [4], stainless steel [5,6], 9% nickel steel [7], titanium alloys [8], other alloys [[9], [10], [11]], and dissimilar material welding [[12], [13], [14]]. These studies highlight the versatility and adaptability of K-TIG welding in diverse material systems.

Beyond material applications, considerable efforts have been dedicated to advancing K-TIG welding methodologies. Xu and Zhan et al. [15,16] investigated the application of sinusoidal alternating axial magnetic fields in K-TIG welding, revealing improvements in microstructure, increased impact toughness, and reduced heat input. Dong et al. [17] optimized high-frequency pulse waveform current on K-TIG for AISI 444 stainless steel, identifying heat input as a critical factor for achieving full penetration. Wang and Jiang et al. [18,19] enhanced aluminum alloy welding using DP-VPTIG, demonstrating improvements in keyhole stability, microstructure, and mechanical properties. Collectively, these studies underscore the potential of K-TIG welding for improve efficiency, superior mechanical performance, and broader material adaptability.

Numerical simulation plays a crucial role in welding research by revealing underlying mechanisms, optimizing process parameters, and minimizing experimental costs. While substantial progress has been made, most studies have primarily focused on conventional GTAW. For instance, Mougnot et al. [20] demonstrating the Marangoni effect's influence on weld pool dynamics with minimal impact from metal vapour. Wang et al. [21,22]

identifying plasma drag shear stress as the dominant force which driving the molten metal flow under high-current GTAW and illustrating electrode interaction effects in double-electrode GTAW. Xiao et al. [\[23\]](#), [\[24\]](#), [\[25\]](#) investigated magnetic field effects, showing that high-frequency axial magnetic fields enhance arc stability via radial Lorentz forces, while alternating fields induce arc rotation. Their unified model further revealed that an axial magnetic field constricts the arc and increases flow velocity, whereas a transverse magnetic field deflects the arc, creating asymmetric heat flux. These studies underscore the pivotal role of magnetic fields in regulating arc behavior and weld pool dynamics.

Numerical studies on GTAW have extensively investigated arc behavior; however, the presence of keyhole formation in K-TIG welding introduces additional complexities. Most research on keyhole dynamics has focused on plasma arc welding (PAW), which shares similarities with K-TIG in terms of deep penetration and keyhole formation. Professor Shujun Chen's team has made significant contributions to PAW simulations. Pan et al. [\[26,27\]](#) developed coupled models for stationary and variable-polarity PAW, identifying keyhole formation stages and weld pool flow mechanisms. Xu et al. [\[28\]](#) found that keyhole formation compresses the plasma arc, increasing both pressure and velocity. Professor Chuansong Wu's team further advanced PAW modeling, with Li et al. [\[29\]](#) predicting keyhole dimensions and weld pool characteristics, while Sun et al. [\[30\]](#) captured the thermal-mechanical effects in pulsed keyholing. Li et al. [\[31\]](#), [\[32\]](#), [\[33\]](#) revealed that Marangoni shear stress dominates weld pool flow, whereas Jian et al. [\[34,35\]](#) provided unified simulations of keyhole, arc, and weld pool interactions. Additionally, Professor Dongsheng Wu's team explored keyhole formation, energy distribution, and arc behavior in PAW, elucidating keyhole depth dynamics and energy loss mechanisms [\[37,38\]](#).

Advancements in multiphysics simulations of K-TIG welding have provided deeper insights into the process, particularly in heat source modeling, arc characteristics, and weld pool dynamic behavior. Using the SYSWELD platform, Li et al. [\[39\]](#) constructed a hybrid thermal model that simulates K-TIG welding processes for titanium alloys,

achieving precise geometrical and thermal predictions of the molten pool. Battista et al. [40] contrasted the MHS and ITC methods for forecasting welding distortions in Inconel 718, with ITC offering better computational efficiency. Song et al. [41] conducted 2D simulations of K-TIG arcs, showing that keyhole formation reduced the electric potential gradient and raised the anode center temperature. Zhan et al. [42] extended this to a 3D model incorporating keyhole depth variations and external sinusoidal alternating-axis magnetic fields, revealing improved penetration and weld pool dynamics.

Research has also focused on K-TIG weld pool dynamic behavior. Chen et al. [43] demonstrated that arc shear stress concentrates the high-temperature molten metal in the upper of the weld pool. Jiang et al. [44] used high speed imaging and CFD modeling to show that periodic keyhole formation enhances heat transfer and weld stability. Through the implementation of a bimodal distribution model, Wang et al. [45] established predictive capabilities for weld pool deformation characteristics, particularly elucidating the combined effects of arc pressure and plasma shear forces. Cui et al. [46] investigated keyhole dynamics, showing that electromagnetic forces, surface tension, and arc pressure regulate keyhole formation and weld penetration, with lower welding speeds enhancing molten metal flow.

In K-TIG welding, high currents (≥ 300 A) generate elevated arc temperatures, leading to base metal (BM) melting and metal vaporization, which significantly influence arc properties. However, most studies on metal vapour focus on conventional TIG welding. A three-dimensional numerical model analyzing arc-weld pool interactions was developed by Fan et al. [47], with results indicating predominant localization of metal vapour near the weld pool region, which induces arc constriction while exhibiting negligible alterations to flow velocity and interfacial shear stress. Xiao et al. [48] extended this to stationary A-TIG welding, demonstrating that higher weld pool temperatures increase metal vapour concentration, reducing penetration depth by 6%. Wang et al. [49,50] analyzed metal vapour behavior in DE-TIG welding, revealing its role in constricting the arc and slightly increasing weld pool temperature.

Further studies have explored metal vapour effects under external influences. Chen et al. [51] examined how magnetic fields alter arc morphology, voltage, and temperature. Murphy et al. [[52], [53], [54]] reviewed metal vapour effects in arc welding, highlighting its impact on penetration and geometry across various processes. Xiang et al. [55] and Iwao et al. [56] investigated metal vapour transport, showing that it reduces arc temperature and current density, leading to shallower weld pools. Tanaka et al. [57], revealing that metal vapour from the molten anode of GTAW modifies arc temperature, current density, and heat source characteristics. These studies underscore the necessity of incorporating metal vapour effects into welding models for accurate predictions.

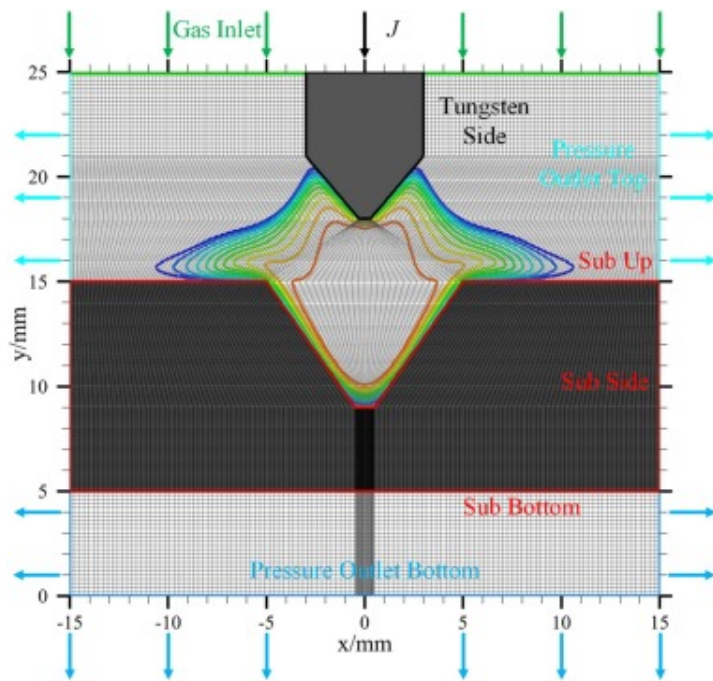
Previous studies have demonstrated that metal vapour significantly influences welding arc characteristics, while keyhole formation plays a crucial role in K-TIG welding. However, research on these two aspects has largely been conducted independently, with limited studies examining their interaction. Given the high currents and deep penetration of K-TIG welding, metal vapour and keyhole dynamics are expected to jointly affect arc behavior and weld formation. Through multi-physics modeling and experimental validation, the work systematically examines the interdependent relationships between arc characteristics, keyhole stability thresholds, and weld pool convection patterns. The derived process – structure – property correlations provide actionable guidelines for enhancing K-TIG welding performance in thick-section metal joining applications.

2. Mathematical model

2.1. Geometry model

Due to the unique characteristics of K-TIG welding, which distinguish it from conventional GTAW methods, it relies on strong arc pressure to penetrate the BM and form a through - thickness keyhole. The arc plasma tail flame emerges from the backside of the BM, enabling single - sided welding and double - sided formation. Conventional GTAW - based simulation models fail to accurately model the distribution of keyhole and tail flame in K-TIG welding, therefore, an improvement is introduced in this study by

incorporating a new gas domain on the bottom of the BM domain to accommodate K-TIG welding, as depicted in Fig. 1.



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Fig. 1. K-TIG welding computational domain diagram.

As there is no additional external magnetic field in this study, therefore, the arc plasma does not undergo rotational or oscillatory motion. Under these conditions, the physical behavior of the arc plasma and its interaction with the keyhole can be reasonably approximated using a 2D axisymmetric model to simplify calculations [28,32,41], and the detailed model setup is the same as in our previous research [42]. The vertical gap separating the tungsten electrode tip from the BM's upper surface was maintained at 3 mm.

A structured mesh is employed in this study. The localized areas adjacent to both the tungsten electrode tip and BM's top surface demonstrate significant thermal gradient fluctuations, finer mesh elements are applied in these areas, while coarser mesh elements are used in other regions to accelerate computation. The specific K-TIG welding simulation boundary conditions are detailed in our previous research [42].

2.2. Basic assumptions

Since K-TIG welding is a highly transient and nonlinear process involving multiple physical fields and complex physicochemical interactions, a reasonable mathematical model is required to enhance computational efficiency. Axisymmetric numerical modeling incorporating tungsten electrode – arc - gas phase interactions (top/bottom domains) with the BM substrate was established under following assumptions.

For the arc plasma domain:

- (1) The arc plasma is assumed to be a Newtonian fluid that remains in a laminar state and is subjected to standard atmospheric pressure. Since the transient behavior of gas phase flow is considered, an unsteady - state calculation is applied.
- (2) The local thermodynamic equilibrium (LTE) assumption is applied, implying that temperature differences between electrons and heavy ions are neglected, and a single particle temperature is used to characterize the system.
- (3) Radiation calculations are expressed using the net emission coefficient, with the reabsorption of radiation energy by the arc's low - temperature regions omitted.
- (4) The thermal losses caused by the viscous effects of the arc plasma are neglected.

- (5) The thermophysical properties of the arc plasma are regarded as functions that depend only on temperature. In the case of including metal vapour, these properties are dealt with as functions related to both temperature and the mass fraction of the metal vapour. A bilinear interpolation method based on temperature and mass fraction is used for calculations.

For the BM domain:

- (6) The molten phase within the BM is governed by Newton's viscosity law, with stress - strain relationships maintaining linear proportionality under subcritical Reynolds number conditions. Since the transient behavior of liquid metal flow is considered, an unsteady - state calculation is applied.
- (7) The BM upper surface is defined as a preset keyhole profile under different keyhole depths, without consideration of thermal deformation or mass loss. Additionally, it is assumed that the metal vapour evaporating from the surface of the high - temperature weld pool, specifically above the gas - phase transition region, is entirely composed of Fe metal vapour, as the BM is Q345 low - carbon steel.
- (8) Thermal expansion effects in buoyancy - driven flows are modeled using the boussinesq density variation hypothesis.

Although the static boundary assumption does not capture dynamic behaviors such as keyhole oscillation or collapse, this simplification is widely adopted in arc plasma simulation studies to reduce computational cost and to isolate the effect of arc plasma transport mechanisms under representative geometric conditions [28, 32, 41, 42, 75]. It enables stable evaluation of arc plasma characteristics across various keyhole stages without introducing the numerical instabilities associated with moving - boundary simulations. Therefore, while the model does not account for fluid deformation, it

remains a feasible and validated approach for exploring arc plasma behavior in relation to static keyhole geometries.

2.3. Governing equations

Under the assumptions outlined above, the model must satisfy the conservation equations of mass, momentum, and energy. Given that the impact of metal vapour is taken into account, the conservation equations for the mass fraction of metal vapour are likewise resolved. This enables the acquisition of the multiphysics distribution traits regarding temperature, velocity, and pressure within the arc plasma.

Mass conservation equation:

$$\frac{\partial(\rho)}{\partial t} + \frac{1}{r} \frac{\partial(r\rho v)}{\partial r} + \frac{\partial(\rho u)}{\partial z} = 0 \quad (1)$$

Momentum conservation equation:

In the axial direction of the model:

$$\begin{aligned} \frac{\partial(\rho u)}{\partial t} + \frac{1}{r} \frac{\partial(r\rho v u)}{\partial r} + \frac{\partial(\rho u u)}{\partial z} = & -\frac{\partial p}{\partial z} + \frac{\partial}{\partial z} \left(2\mu \frac{\partial u}{\partial z} \right) + \frac{1}{r} \frac{\partial}{\partial r} \left[r\mu \left(\frac{\partial u}{\partial r} + \frac{\partial v}{\partial z} \right) \right] \\ & + j_r B_\theta + \rho g + G + S_u \end{aligned} \quad (2)$$

where G represents thermal buoyancy, which is described using the boussinesq approximation:

$$G = -\rho g \beta (T - T_{\text{ref}}) \quad (3)$$

In the radial direction of the model:

$$(4)$$

$$\frac{\partial \rho v}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r \rho v v) + \frac{\partial}{\partial z} (\rho u v) = -\frac{\partial p}{\partial r} + \frac{1}{r} \frac{\partial}{\partial r} \left(2r\mu \frac{\partial v}{\partial r} \right) + \frac{1}{r} \frac{\partial}{\partial z} \left[r\mu \left(\frac{\partial u}{\partial r} + \frac{\partial v}{\partial z} \right) \right] - 2\mu \frac{v}{r^2} - j_z B_\theta + S_u$$

Energy conservation equation:

$$\frac{\partial \rho T}{\partial t} + \frac{1}{r} \frac{\partial (\rho c_p v T)}{\partial r} + \frac{\partial (\rho c_p u T)}{\partial z} = \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) + S \quad (5)$$

In the above equations, u and v denote the axial (z) and radial (r) velocity components, respectively. The arc plasma pressure is given by p , and the gravitational acceleration is represented by g . The parameters ρ , μ , c_p , k , and S correspond to the density, viscosity coefficient, specific heat capacity, thermal conductivity, and source term in the energy equation for argon gas, respectively. β (10^{-4}) is used to represent the coefficient of thermal expansion of BM.

Considering the impact of metal vapour, a species transport equation is introduced to manage the transport of metal vapour. In the BM, there are metal elements like Cu, Mn, Al, and Fe, as shown in Table 1. To account for the influence of the metal elements in BM, the species mass conservation equations for individual species have to be activated, as follows [58]:

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \underline{v} Y_i) + \nabla \cdot \underline{J}_i = r_i \quad (6)$$

Table 1. Chemical composition of Q345 base metal [4].

Element	C	Si	Mn	P	S	Al	Cu	Fe
Content	≤0.2	≤0.55	1.20–1.70	≤0.025	≤0.012	≥0.02	≤0.30	Bal.

Here, Y_i denotes the mass fraction of species i , and r_i represents the net production rate of species i resulting from chemical reactions. In a gas or plasma composed of N different

species, the mass diffusion flux of species i can be expressed as:

$$\underline{J}_i \equiv m_i n_i \underline{v}_i = \frac{m_i n^2}{\rho} \sum_{j=1}^N m_j D_{ij} \underline{d}_j - D_i^T \nabla \ln T \quad (7)$$

In this context, $\underline{v}_i$ represents the diffusion velocity of species i relative to the mass - averaged flow, while m_i and n_i correspond to the mass and number density of species i , respectively. The ordinary diffusion coefficients D_{ij} and thermal diffusion coefficients D_i^T are multicomponent transport properties. Determining these values requires knowledge of the mole fractions, molecular masses of all species involved, and the binary collision cross - sections for each species pair. The associated driving force term, $\underline{d}_j$, is defined as:

$$\underline{d}_j = \nabla x_j + \left(x_j - \frac{n_j m_j}{\rho} \right) \nabla \ln P - \frac{n_j m_j}{P \rho} \left(\frac{\rho}{m_j} \underline{F}_j - \sum_{l=1}^N n_l \underline{F}_l \right) \quad (8)$$

The three components correspond to the effects of mole fraction gradients x_j , pressure variations P , and external forces acting on species $\underline{F}_j$, respectively. Because the diffusion velocities are measured relative to the mass - averaged flow, it is sufficient to determine the velocities for only $N - 1$ species in a system containing N components. Therefore, the conservation equation (Eq. (6)) needs to be solved for just $N - 1$ individual species.

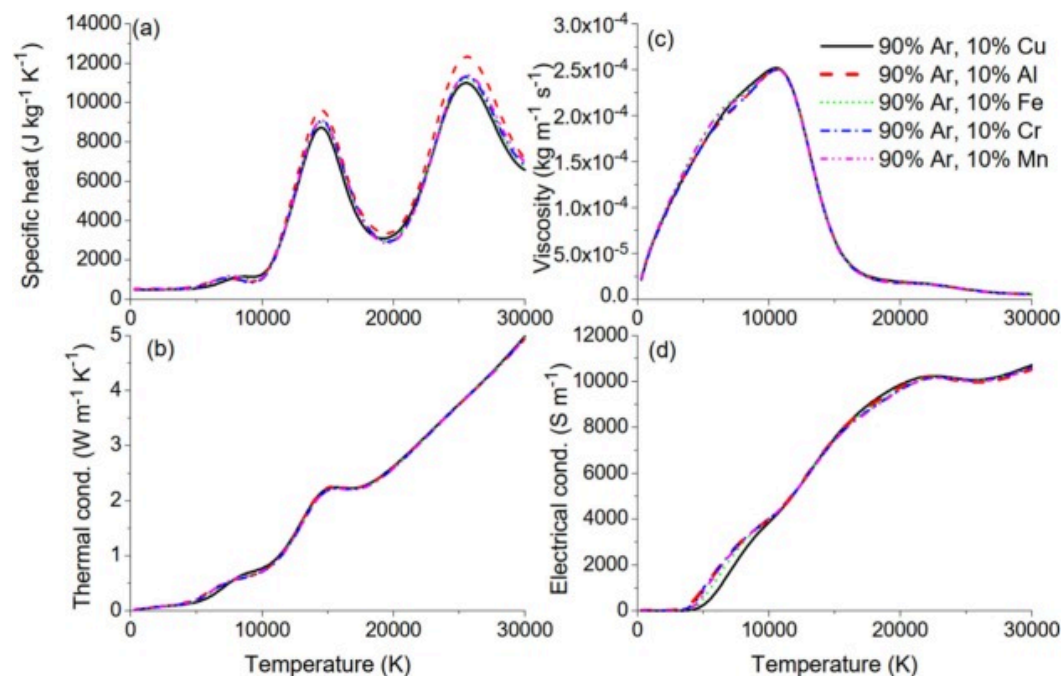
Eqs. (6), (7), (8) apply to both neutral gases and ionized gases such as plasmas. However, for plasmas, the determination of diffusion coefficients must include the effect of ambipolar diffusion. This phenomenon occurs because electrons, having significantly lower mass than ions, tend to diffuse faster, leading to charge separation. As a result, an electric field – referred to as the ambipolar electric field – is generated. This field acts to reduce the diffusion rate of electrons while simultaneously accelerating that of the ions. To incorporate this effect, the expression for the external force term in Eq. (8) is modified accordingly.

$$\underline{F}_i = e Z_i (\underline{E} + \underline{E}^a) \quad (9)$$

This approach accounts for both the externally applied electric field $\underline{E}$ and the ambipolar electric field $\underline{E}^a$, where Z_i represents the charge number of species i . The ambipolar field can be incorporated in two ways: either explicitly, by directly substituting Eq. (9) into Eq. (8), or implicitly, by adjusting the multicomponent diffusion coefficients to embed its influence. When the latter strategy is adopted, Eq. (9) includes only the externally imposed electric field.

Based on the preceding analysis, determining species diffusion involves computing binary diffusion coefficients for every species pair along with individual thermal diffusion coefficients, which entails a substantial computational burden. Therefore, given the minor influence of trace alloying elements and the high computational costs involved, we adopted a simplified approach in this study that considers only Fe vapour in the simulation [25,48], and is further justified by Murphy's [52,58] comprehensive analysis, which demonstrated the following:

1. Various metal vapours, such as Fe, Cu, and Mn, exhibit qualitatively similar trends in their influence on arc plasma properties - including viscosity, specific heat, thermal conductivity, and electrical conductivity - as shown in Fig. 2. This observation supports the grouping of less abundant metallic species into a single representative vapour (Fe), which is a reasonable and widely accepted simplification.



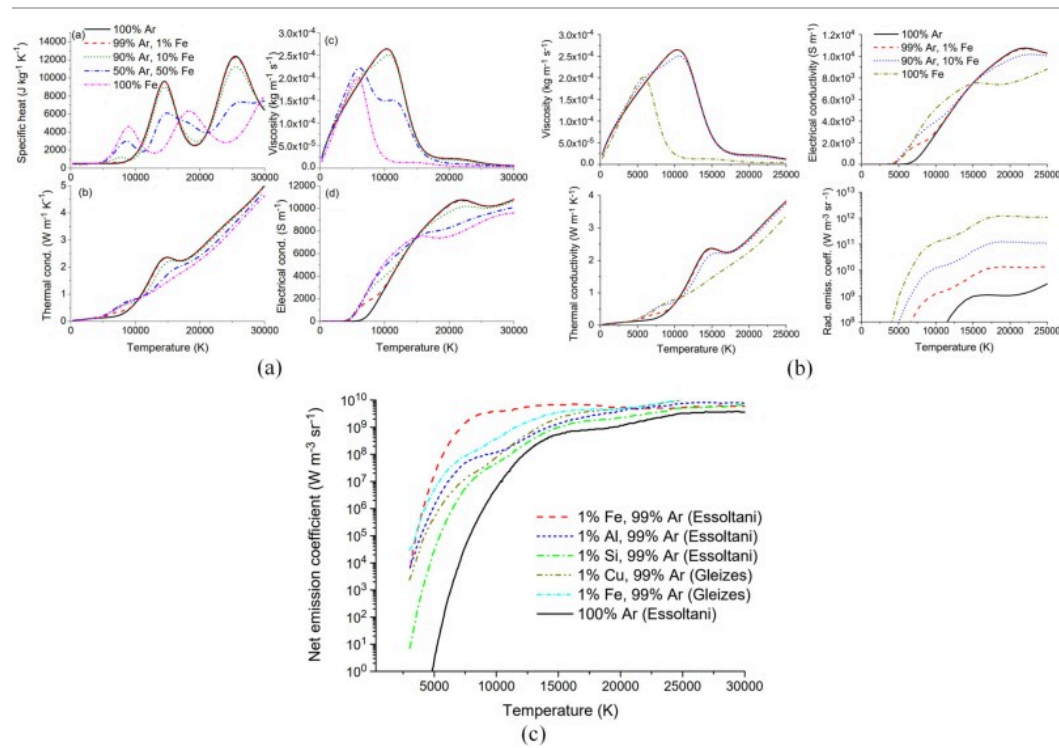
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Fig. 2. Calculated properties of arc plasmas in mixtures of 90% argon and 10% metal.

vapour by mole, for five different metals [52].

- As illustrated in Fig. 3, even small variations in Fe concentration cause significant changes in the arc plasma's thermal physical properties. This confirms that Fe vapour plays a dominant role in governing arc plasma characteristics, while other metal species have only a marginal impact due to both their lower content and weaker thermophysical influence.



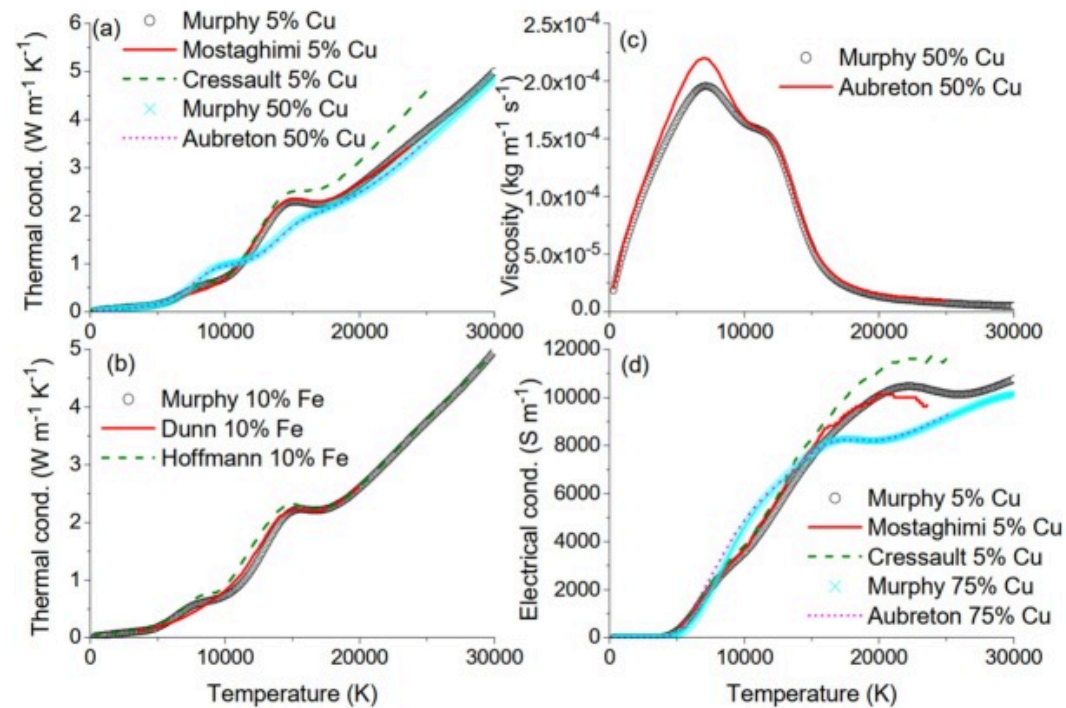
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Fig. 3. (a) Calculated properties of plasmas in different mixtures of argon and iron vapour percentages are by mole (b) Dependence of viscosity, electrical conductivity, thermal conductivity and net radiative emission coefficient on temperature for argon, iron vapour, and mixtures of argon with 1 mol% and 10 mol% iron vapour, at 1 atm (c) Comparison of the net emission coefficients for mixtures of argon with 1% by mole of iron, copper, aluminum and silicon vapours, and pure argon. The plasma radius is 1 mm [52,58].

3. Furthermore, Fig. 4 shows that consistent arc plasma property data have been obtained across different studies using Murphy's theoretical

framework, validating the accuracy and general applicability of the model used in this work.



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Fig. 4. Comparison of transport properties of argon copper and argon iron plasmas calculated by different authors. Percentages are by mole. (a) Thermal conductivity of argon copper mixtures, (b) Thermal conductivity of an argon - iron mixture, (c) Viscosity of an argon copper mixture, (d) Electrical conductivity of argon copper mixtures [52].

Under this binary gas assumption, and in line with the LTE framework, individual species are grouped into broader categories. For instance, in an argon - based plasma containing iron vapour, all argon - related particles (Ar , Ar^+ , Ar^{2+} , Ar^{3+} , etc.) are classified under the

argon group, while Fe, Fe⁺, Fe²⁺, Fe³⁺, and similar species are grouped as iron vapour.

Electrons are partitioned between the groups to preserve overall charge neutrality. With this grouping strategy, the $N - 1$ conservation equations typically required for each individual species (e.g., Eq. (6)) can be condensed into a single conservation equation for each gas, as illustrated by Eq. (10) [58]:

$$\frac{\partial(\rho Y_m)}{\partial t} + \nabla(\rho Y_m v) = \nabla(\rho D \nabla Y_m) \quad (10)$$

where, Y_m denotes the mass - based fraction of Fe metal vapour, and D represents the coefficient of diffusion for Fe metal vapour.

The binary diffusion coefficient D of the Fe metal vapour in Eq. (10) is approximated using the second viscosity approximation [58,59], and its expression is given by:

$$D = \frac{2\sqrt{2}(1/M_1 + 1/M_2)^{0.5}}{\left\{ (\rho_1^2/\beta_1^2\mu_1^2M_1)^{0.25} + (\rho_2^2/\beta_2^2\mu_2^2M_2)^{0.25} \right\}^2} \quad (11)$$

In the equation, M_1 denote the molar masses of Fe metal vapour, and M_2 denote the molar masses of pure argon shielding gas. ρ_1 and η_1 represent the density and viscosity of Fe metal vapour, respectively, while ρ_2 and η_2 correspond to those of pure argon shielding gas. β_1 and β_2 are dimensionless constants obtained from neutral atom collision calculations. For a wide range of different gases, these values typically range from 1.20 to 1.543, and a value of 1.385 is adopted in this study [57,60,61].

Metal vapour concentration is constrained to zero value at all cathodic interfaces and solid BM regions. In regions of the anode surface where the temperature surpasses the boiling point, the metal vapour mass fraction Y_m is calculated using the saturated vapour pressure formula:

$$Y_m = \frac{P_v(T)M_1}{P_v(T)M_1 + (P_0 - P_v(T))M_2} \quad (12)$$

In the equation, P_0 is the fixed atmospheric reference pressure (1.01325×10^5 Pa), $P_v(T)$ quantifies the volatile metal vapour pressure as a function of local temperature, which is

expressed as follows [58,62]:

$$P_v(T) = P_0 \exp \left[\frac{-H_{\text{vap}}}{R} \left(\frac{1}{T} - \frac{1}{T_b} \right) \right] \quad (13)$$

H_{vap} is the molar heat of evaporation ($349.6 \text{ kJ} \cdot \text{mol}^{-1}$), T_b is the boiling temperature of the BM at atmospheric pressure (3050K) and R is the ideal gas constant ($8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$).

On the upper surface of the weld pool, the theoretical range of the metal vapour concentration Y_m is from 0 to 1.0. On the symmetry axis, $\partial Y_m / \partial x = 0$, and for the other boundaries, the concentration value is set to 0.

Current continuity equation:

$$\sigma \frac{\partial \varphi}{\partial t} + \frac{\partial}{\partial z} \left(\sigma \frac{\partial \varphi}{\partial z} \right) + \frac{1}{r} \frac{\partial}{\partial r} \left(r \sigma \frac{\partial \varphi}{\partial r} \right) = 0 \quad (14)$$

Ohm's law:

$$j_z = -\sigma \frac{\partial \varphi}{\partial z} \quad (15)$$

$$j_r = -\sigma \frac{\partial \varphi}{\partial r} \quad (16)$$

Self - consistent magnetic field computation in the arc plasma column requires solving the magnetohydrodynamic induction equation, axial magnetic vector A_z and radial magnetic vector A_r are introduced, which satisfy the following conditions:

$$\frac{\partial A_z}{\partial t} + \frac{\partial}{\partial z} \left(\frac{\partial A_z}{\partial z} \right) + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial A_z}{\partial r} \right) = -\mu_0 j_z \quad (17)$$

$$\frac{\partial A_r}{\partial t} + \frac{\partial}{\partial z} \left(\frac{\partial A_r}{\partial z} \right) + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial A_r}{\partial r} \right) = -\mu_0 j_r + \frac{A_r}{r^2} \quad (18)$$

where, φ is the electric potential, μ_0 is the vacuum permeability.

By introducing A_z and A_r , the circumferential component of the arc's self - induced magnetic field can be calculated:

$$B_\theta = \frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \quad (19)$$

The axial and radial current densities are denoted as j_z and j_r , while B_θ represents the circumferential component of the self - induced magnetic field of the arc plasma. The temperature is represented by T , and the ambient temperature is set as $T_0=300\text{K}$.

In the aforementioned conservation equations, certain processes cannot be described solely by the unsteady term, convection term, and diffusion term. Therefore, source terms are introduced to account for these effects. For the solid - liquid phase transition of the BM, we used our previous method of adding a source phase S_u to the momentum conservation equation [42]. The expression for S_u is as follows:

$$S_u = A_{\text{mush}} \frac{(1-f_1)^2}{f_1^3 + \delta} (v \cdot v_p) \quad (20)$$

Here, A_{mush} is set as a constant (10^5), f_1 represents the liquid volume fraction, δ is 0.0001, a small value. v represents the fluid velocity vector, while v_p stands for the drag velocity. It is assumed that the liquid volume fraction f_1 varies linearly with the temperature of the BM, as shown in the following expression:

$$f_1 = \begin{cases} 0 & T \leq T_s \\ \frac{T - T_s}{T_1 - T_s} & T_s < T \leq T_1 \\ 1 & T > T_1 \end{cases} \quad (21)$$

In the arc plasma region, since without phase change between solid and liquid, the momentum source term S_u is set to zero.

S is the energy source term, represented in the arc plasma region as:

$$S = \frac{j_r^2 + j_z^2}{\sigma} + \frac{5k_B}{2e} \left(j_r \frac{\partial T}{\partial r} + j_z \frac{\partial T}{\partial z} \right) - S_R \quad (22)$$

$$S_R = 4\pi\epsilon_n \quad (23)$$

S_R can also be expressed as a temperature - dependent function.

$$S_R = \varepsilon \sigma_0 (T^4 - T_0^4) \quad (24)$$

The Eq. (22) includes Joule heating, electronic transport enthalpy, and arc plasma radiation heat loss. Here, the Boltzmann constant is given by $k_B = 1.38 \times 10^{-23} \text{ J} \cdot \text{K}^{-1}$, e represents the electronic charge, which is set to $1.602 \times 10^{-19} \text{ C}$ in this model, and ε_n is the net radiation heat loss coefficient considering the Ar - Fe mixture ratio, which is obtained through a binary linear interpolation based on temperature and metal vapour mass fraction. ε is the radiation coefficient of the arc plasma ($0.4 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}$). σ_0 is Stephen Boltzmann constant ($5.67 \times 10^{-8} \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-4}$), and T_0 is ambient temperature (300K).

For the weld pool, only the latent heat of melting is considered, as the Joule heating effect in the weld pool is relatively minor [[63], [64], [65]]. Therefore, the effects of the unsteady term and the convective term on phase change latent heat are taken into account:

$$S = \frac{\partial \rho f_1 L}{\partial t} + \nabla \cdot (\rho v f_1 L) \quad (25)$$

L represents the latent heat of melting of BM.

2.4. Calculation boundary conditions

The applied boundary conditions in all computational domains maintain consistency with our prior established numerical model [42]. The 500 A welding current condition is applied, while anode/cathode boundary configurations replicate the validated procedures from prior works [[66], [67], [68]].

2.5. Material thermal properties

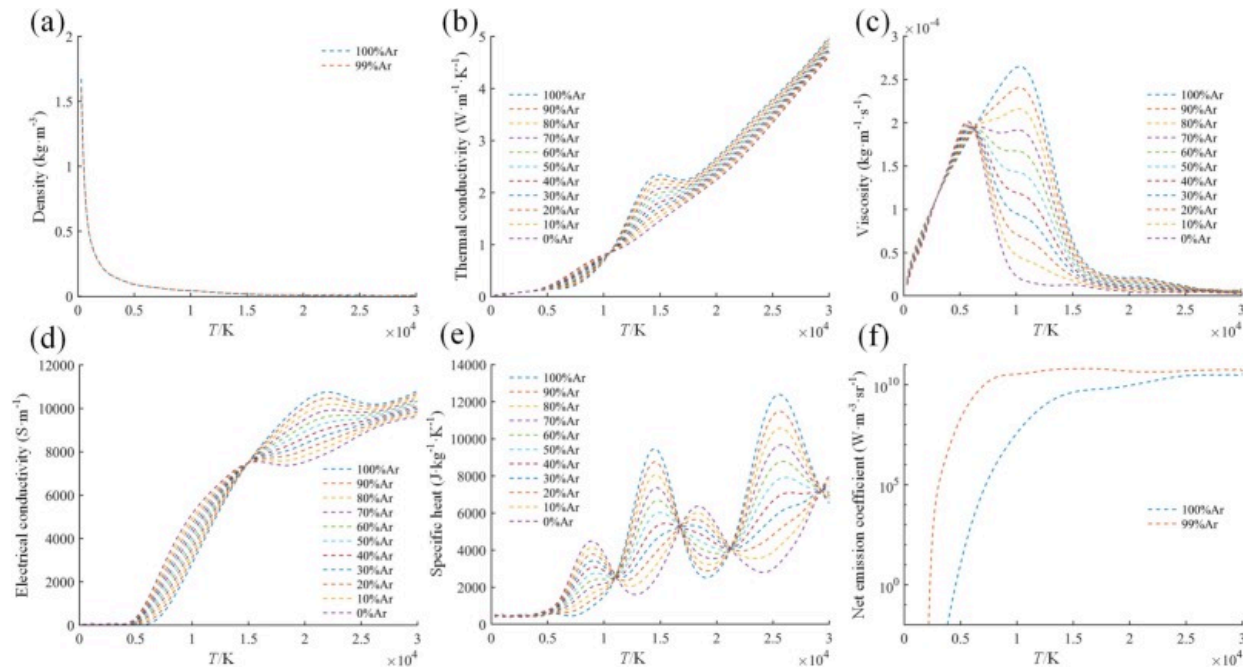
The numerical simulation adopts thermophysical parameters for the arc plasma from Refs. [50,52,54,[69], [70], [71], [72], [73], [74]]. These parameters are pre - calculated and structured into two - dimensional arrays, with rows corresponding to discrete Fe mass

fractions (ranging from 0 to 1.0 in increments of 0.1) and columns corresponding to discrete temperatures (from 300K to 30,000K, incremented every 100K).

During the simulation, properties at intermediate conditions are efficiently interpolated as follows:

1. Identify the mass fraction row: Determine the row in the array corresponding to the current computational node's mass fraction.
2. Identify bounding temperature columns: Determine the two adjacent columns (temperatures) in the array that bracket the node's actual temperature.
3. Perform linear interpolation in temperature: Calculate the property value at the node by performing linear interpolation between the identified bounding temperature values, proportional to the node's actual temperature.

This interpolation approach provides accurate and efficient access to the necessary thermophysical properties at any intermediate combination of temperature and plasma composition. Specific examples of these interpolated properties, including density, thermal conductivity, and net radiation emission coefficients, are depicted in [Fig. 5](#), while thermophysical properties of the Q345 BM are presented separately in [Table 2](#).



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Fig. 5. Thermal properties of arc plasma.

Table 2. Thermophysical properties of Q345 base metal.

Physical property	Value
Density $\rho/(\text{kg}\cdot\text{m}^{-3})$	7200
Electrical conductivity $\sigma/(\text{S}\cdot\text{m}^{-1})$	7.7×10^5
Thermal conductivity $k/(\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1})$	[52, 72]
Viscosity $\mu/(\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-1})$	[52, 72]
Boltzmann constant $k_B/(\text{J}\cdot\text{K}^{-1})$	1.38×10^{-23}

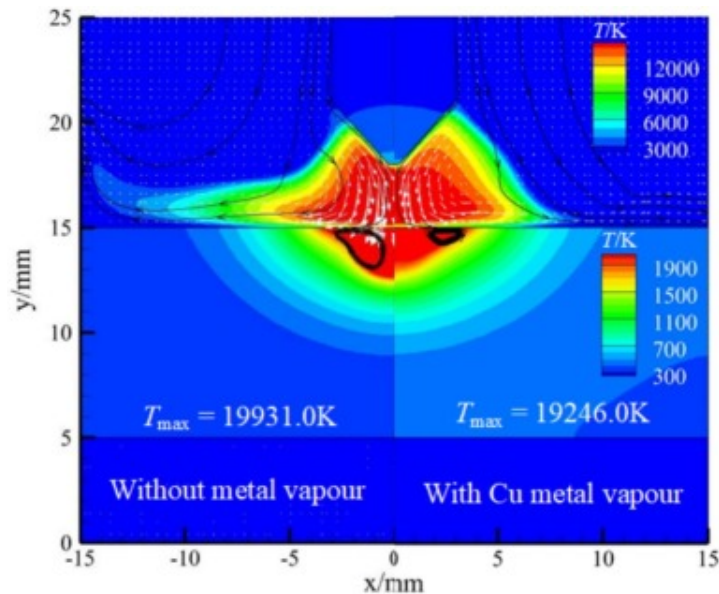
Physical property	Value
Stefan Boltzmann constant $\sigma_s/(\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-4})$	5.67×10^{-8}
Vacuum permeability $\mu_0/(\text{H}\cdot\text{m}^{-1})$	$4\pi \times 10^{-7}$
Ambient temperature T_{ref}/K	300
Latent heat of fusion $L/(\text{J}\cdot\text{kg}^{-1})$	2.47×10^5
Liquidus Temperature T_l/K	1793
Solidus temperature T_s/K	1700
Wall radiation heat dissipation coefficient ε_0	0.4
Convective heat transfer coefficient $h_c/(\text{W}\cdot\text{m}^{-2})$	80
Thermal expansion coefficient β	10^{-4}

2.6. Solution setting

The calculation setup is consistent with our previous research [42], with the exception that the algorithm has been modified to SIMPLEC. First, the upper surface of the BM is defined as an adiabatic wall, and steady - state calculations are performed for 10,000 iterations to achieve steady - state conditions for the arc plasma temperature field, velocity field, and pressure field. Afterward, the anode boundary condition is applied, and the species transport equations are activated, transitioning to unsteady - state calculations. The time step is set to 5e^{-4} s, with 50 iterations per time step. It takes 8h to obtain the physical field results for 500ms of simulation on a system with a 10-core intel CPU.

3. Results and discussion

To verify the accuracy of the above analysis regarding the influence of trace elements such as Cu, Mn, and Al on the arc plasma, a preliminary simulation study was conducted. Specifically, we compared two cases: one without considering metal vapour and another considering only Cu metal vapour. The resulting temperature field distributions are shown in Fig. 6. Although the presence of Cu metal vapour does affect the arc plasma temperature field – reducing the peak temperature from 19,931.0K to 19,246.0K – the reduction is relatively minor. Moreover, as shown in Fig. 6, the weld pool depth in the BM region decreases slightly when Cu vapour is included, but the overall contour remains largely unchanged. These results indicate that the influence of trace metal vapours such as Cu is limited. Therefore, to balance computational cost and modeling complexity, their effects can be reasonably neglected in the current study.



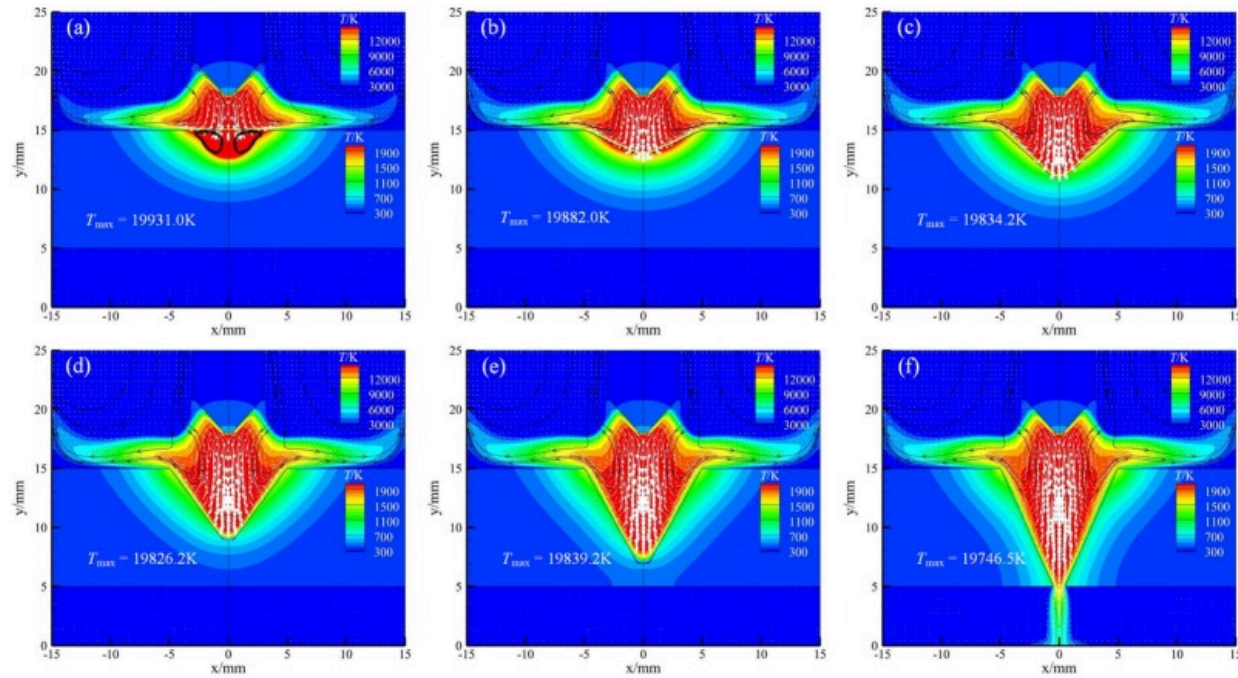
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Fig. 6. Effect of metal vapour on arc plasma.

3.1. Effect of Fe metal vapour on arc plasma temperature field distribution

Fig. 7 displays the temperature and flow field pattern of the K-TIG arc plasma at 500ms under various keyhole depths, without the effect of metal vapour. To mitigate inaccuracies arising from the arc plasma not reaching steady state within the short simulation time, this study focuses on the steady - state arc plasma results at 500ms. As displayed in Fig. 7, with the keyhole depth increasing, the high - temperature area of the arc plasma stretches and penetrates further into the keyhole. This shift contributes to heating the BM, enhancing thermal energy utilization and improving welding efficiency. When the keyhole depth is 0mm, the highest temperature of the arc plasma reaches 19,931.0K, as depicted in Fig. 7(a), and the BM domain above the liquidus temperature is relatively large. As the keyhole forms, the highest temperature of the arc plasma slightly decreases. This occurs because the increasing keyhole depth enables the arc plasma to gain entry into the keyhole, thereby increasing the contact area with the BM domain. Consequently, the BM absorbs more heat, resulting in a reduction in the arc plasma temperature, as depicted in Fig. 7(b) - (e). Notably, when the keyhole completely penetrates the BM, a portion of the arc plasma exits from the bottom side of the BM, removing significant heat. Concurrently, due to the expansion of the arc plasma - BM interaction zone, more heat is conducted into the BM. As a result, the maximum temperature drops to 19,746.5K, as depicted in Fig. 7(f).



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Fig. 7. Temperature field (without metal vapour).

From Fig. 7(a)–(e), it is evident that when the keyhole depth is 0 or 2 mm, the BM domain still exhibits a weld pool with temperatures above the liquidus temperature at 500 ms. However, as the keyhole depth increases further, the temperature in the BM domain falls below the liquidus temperature. This phenomenon can be explained by the following: while the formation of the keyhole increases the contact area between the arc plasma and the BM, it also results in a larger portion of the BM directly exchanging heat with the arc plasma. In other words, the same amount of heat is now distributed over a larger volume of BM, leading to a reduction in the heat energy per unit volume. As a result, the liquidus temperature is no longer reached. To observe temperatures above the liquidus temperature in the BM domain, a longer simulation time would be required.

Furthermore, as indicated by the black trajectory lines of the arc plasma in Fig. 7(a), the arc plasma flows downward. Upon reaching the BM surface, the flow direction is altered by BM obstruction, resulting in horizontal outward movement along the BM surface. As depicted in Fig. 7(b)–(e), with the increase in keyhole depth, a significant amount of arc plasma flows into the keyhole. Nevertheless, because of the progressive reduction in the keyhole radius from top to bottom, the arc plasma experiences intense mechanical compression. At this stage, the BM is not fully melted. As a result, under the obstruction of the BM, the arc's motion direction gradually shifts and becomes inclined upward along the keyhole wall, ultimately flowing outward along the BM surface.

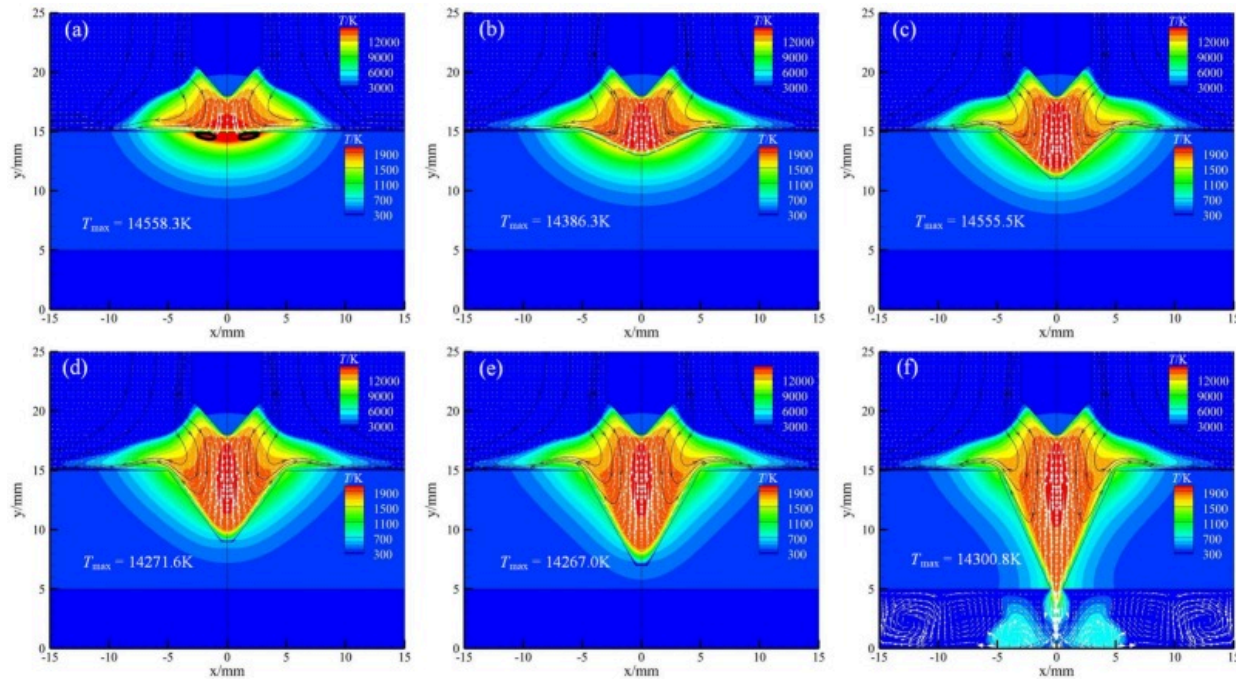
When the BM is completely penetrated, as shown in Fig. 7(f), although partial arc plasma penetration occurs, with some arc plasma emerging from the BM's backside, the majority of the arc plasma continues to move upward along the keyhole wall in an inclined direction. The constrained dimensions of the keyhole exit lead to this phenomenon, ultimately forcing the arc plasma to exit through the BM backside surface.

Fig. 8 displays the temperature and flow field pattern of the arc plasma at 500ms under various keyhole depths, considering the effects of metal vapour. When compared with the results obtained without taking metal vapour into account, one of the most prominent alterations is the substantial reduction in the arc's temperature. At identical color scaling, the arc plasma core region exhibits orange - red coloration in simulations neglecting metal vapour effects, reaching peak temperatures above 19,000K. However, when the metal vapour effect is included, the temperature field of the arc plasma core region becomes light red, with the highest temperature around 14,000K, representing a reduction of approximately 5000K, or a 36% decrease. This phenomenon can be attributed to the following factors:

1. As illustrated in Fig. 5(d), metal vapour exhibits significantly higher electrical conductivity than pure argon shielding gas in the medium - temperature range (5000–15,000K), thereby enhancing the overall electrical conductivity of the arc plasma. According to Eq. (22), an increase in

electrical conductivity leads to a decrease in Joule heating, which consequently results in a substantial reduction in arc plasma temperature.

2. In addition, the presence of metal vapour reduces the arc plasma current density due to the improved electrical conductivity. As shown in [Fig. 20](#) and [Fig. 21](#), the maximum current density decreases by approximately $1.6 \times 10^8 \text{ A} \cdot \text{m}^{-2}$. This reduction further diminishes the Joule heat generation, as described by Eq. [\(22\)](#).
3. Furthermore, a lower current density also changes the electron transport enthalpy, as dictated by Eq. [\(22\)](#), further contributing to the decline in arc plasma temperature.
4. The presence of metal vapour affects the radiation heat loss, as shown in [Fig. 5\(f\)](#), which leads to a decrease in the arc plasma temperature, as described by Eq. [\(22\)](#).



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Fig. 8. Temperature field (with metal vapour).

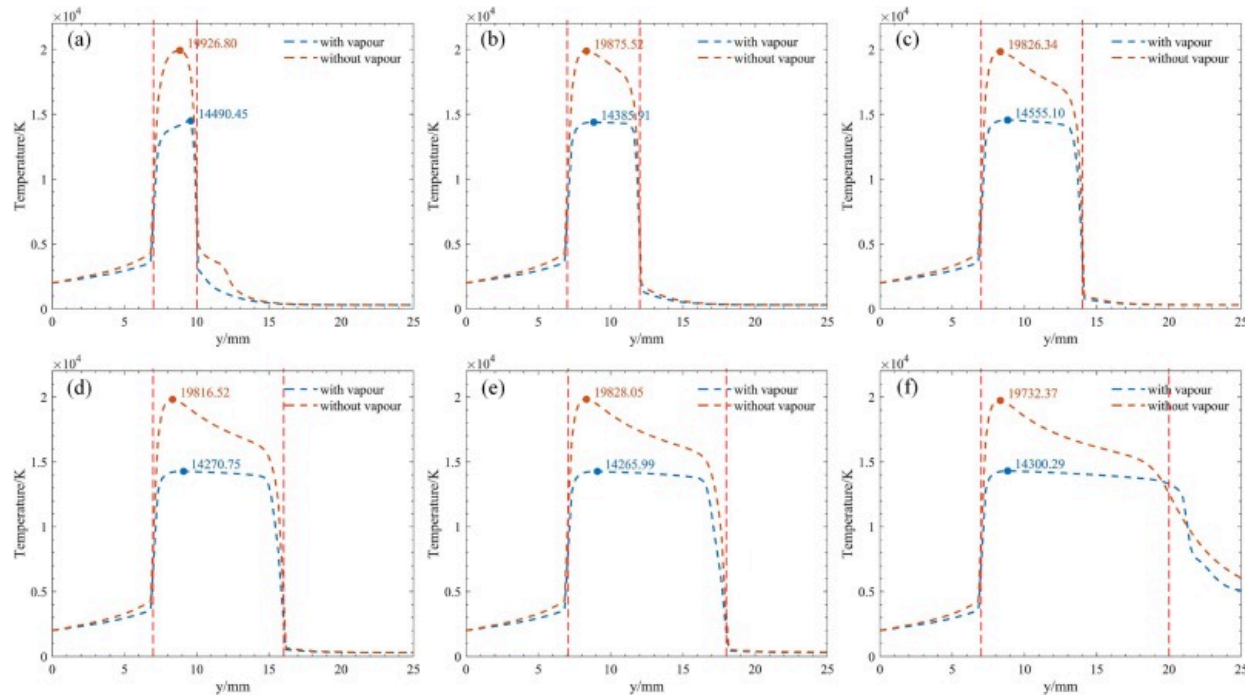
Combined, these four factors lead to a pronounced decrease in arc plasma temperature, particularly within the high - temperature core region. This behavior aligns well with observations reported in previous studies on conventional GTAW processes [25,47,48,58].

Furthermore, from Fig. 8 we can see that the temperature and flow field trends follow the same pattern as when metal vapour is not considered. With the keyhole depth at 0mm, the arc plasma attains its peak temperature of 14,558.3K, and its path exhibits a downward trend, ultimately flowing horizontally outward along the BM's surface. At this point, a weld pool with a temperature above the liquidus line can still be observed in the BM domain, but its volume is noticeably smaller compared to when metal vapour is not considered. The observed phenomenon primarily stems from metal vapour - induced arc

plasma temperature reduction, consequently diminishing thermal transfer from the arc plasma to the BM.

When the keyhole depth reaches 2mm or deeper, no temperature above the liquidus line can be observed in the BM domain. The observed phenomenon occurs because metal vapour lowers arc plasma temperature, and deeper keyholes intensify arc plasma to wall heat transfer. The combined effect of these two factors leads to this observation.

For comparative purposes, the temperature distributions along the axis of symmetry under various keyhole depths were retrieved at 500ms, both with and without considering metal vapour, as depicted in [Fig. 9](#). As previously discussed, the maximum temperature decreases when metal vapour is considered, dropping from 19,926.80K to 14,490.45K when the keyhole depth is 0mm. Consequently, significant differences are observed in the temperature profiles within the arc plasma column region (between the red dashed lines), whereas they nearly overlap in other regions.



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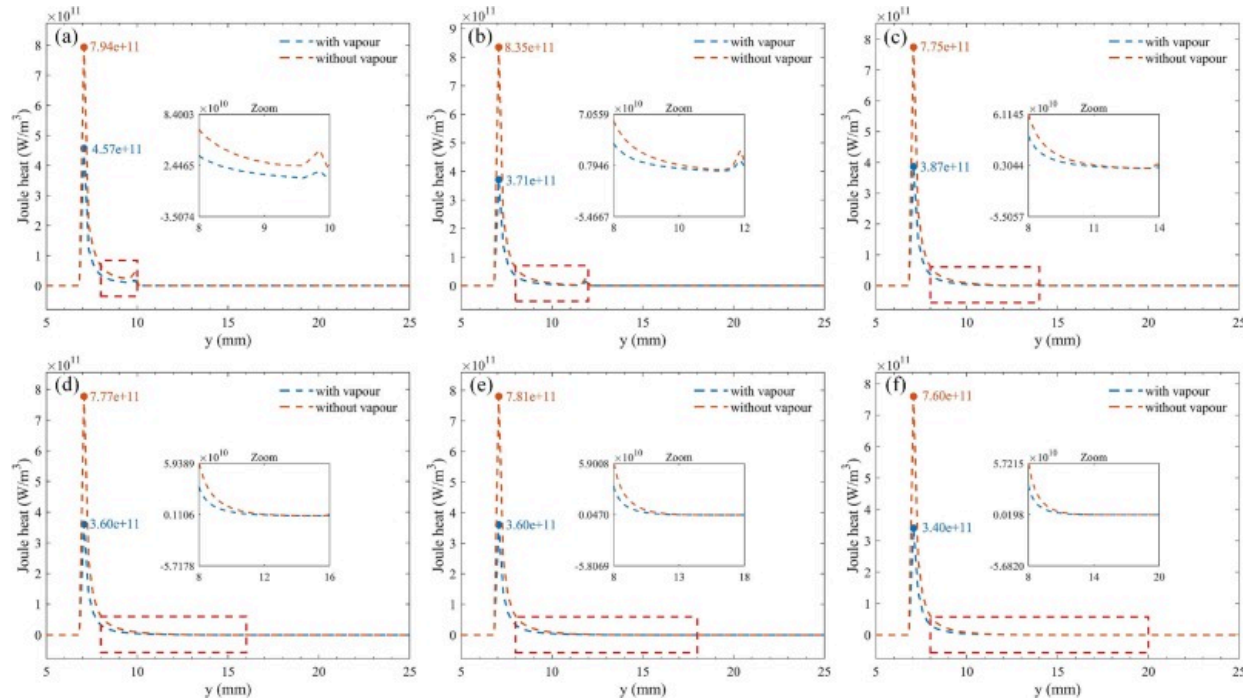
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Fig. 9. Temperature profiles along the symmetry axis.

As depicted in Fig. 9, the temperature in the tungsten electrode region begins at 2000K and gradually rises. After passing the tungsten electrode tip, the temperature rises sharply at an extremely high rate, reaches its maximum (about 19,500K and 14,500K, respectively), and then gradually stabilizes. As it approaches the BM, the temperature rapidly decreases again before continuing to decline at a much slower rate in the BM domain. Additionally, the highest temperature points are consistently located just beneath the tungsten electrode tips. Moreover, in the arc plasma column region, the temperature profile exhibits a gentler gradient when metal vapour is considered, whereas a steeper temperature drop is observed in the absence of metal vapour. To further

investigate this phenomenon, the characteristics of these peak temperature points were extracted for detailed analysis.

To elucidate the underlying cause of the arc plasma temperature reduction when metal vapour is taken into account, the axial distributions of Joule heat, electron transport enthalpy, and thermal radiation loss at various keyhole depths were extracted, as shown in Fig. 10–Fig. 12.



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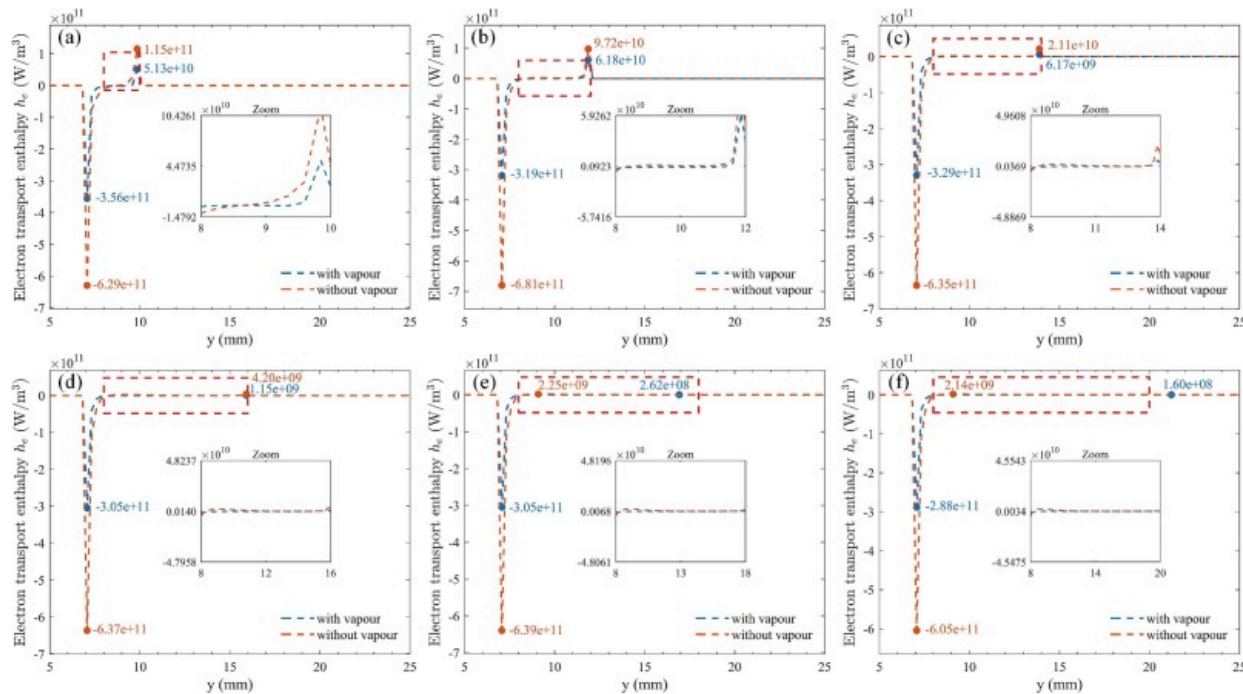
Fig. 10. Joule heat along the symmetry axis.

As illustrated in Fig. 10, and consistent with the preceding analysis, the presence of metal vapour leads to a pronounced decrease in the Joule heat of the arc plasma. When the keyhole depth is 0mm (Fig. 10(a)), the maximum value decreases from $7.94 \times 10^{11} \text{ W} \cdot \text{m}^{-3}$ to $4.57 \times 10^{11} \text{ W} \cdot \text{m}^{-3}$, corresponding to an approximate 50% reduction. With increasing

keyhole depth, the maximum value further decreases to around $3.5 \times 10^{11} \text{ W} \cdot \text{m}^{-3}$, as shown in Fig. 10(b)–(f), and remains predominantly localized near the tungsten electrode tips, where the high current density results in greater Joule heat generation.

In other regions, as revealed by the magnified insets in Fig. 10, the difference in Joule heat between the cases with and without metal vapour progressively diminishes toward the keyhole bottom, eventually converging to nearly identical values.

The variation in electron transport enthalpy along the symmetry axis at different keyhole depths, with and without considering metal vapour, is shown in Fig. 11. It can be observed that at the tungsten electrode tips, the electron transport enthalpy is significantly higher in magnitude than in other regions and exhibits a large negative value. This is because electron transport enthalpy is associated with both the temperature gradient and current density (Eq. (22)). At the tungsten electrode tips, the current density is high (Fig. 19), and the temperature variation is pronounced (Fig. 9), resulting in a greater magnitude of electron transport enthalpy in this region than elsewhere.



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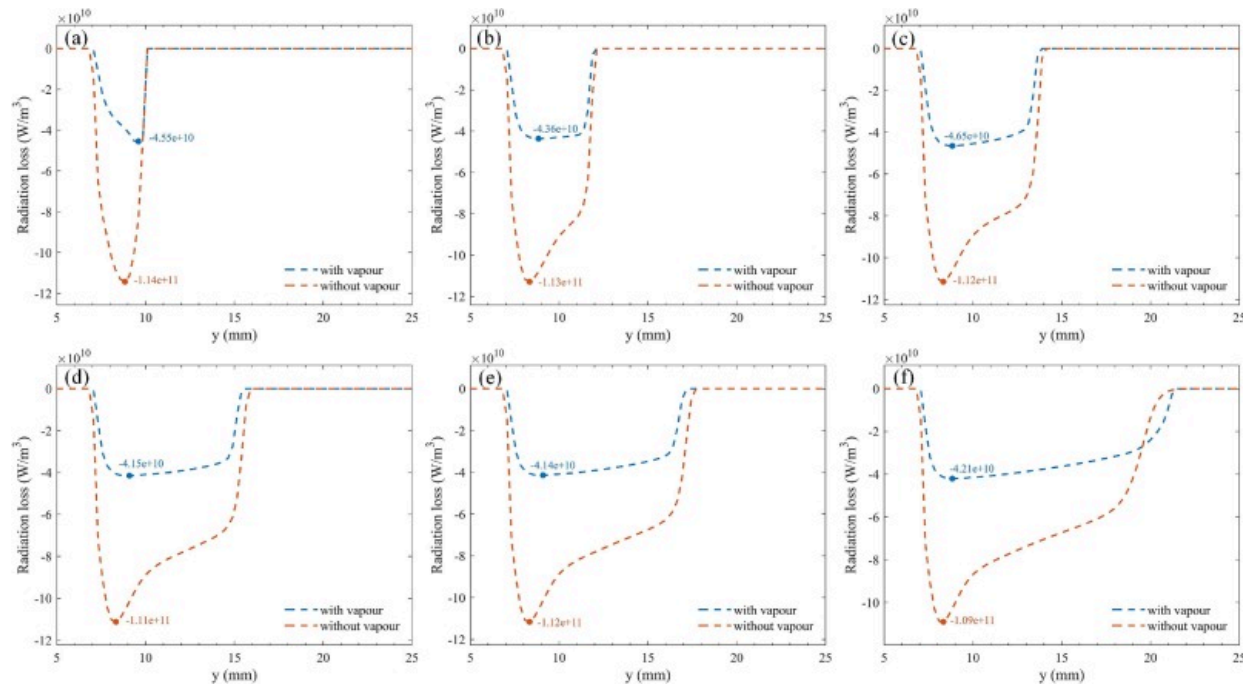
Fig. 11. Electron transport enthalpy along the symmetry axis.

When the keyhole depth is 0 mm (Fig. 11(a)), the maximum electron transport enthalpy decreases from $-6.29 \times 10^{11} \text{ W} \cdot \text{m}^{-3}$ to $-3.56 \times 10^{11} \text{ W} \cdot \text{m}^{-3}$, corresponding to a reduction of $-2.73 \times 10^{11} \text{ W} \cdot \text{m}^{-3}$, approximately halving its magnitude. As the keyhole depth increases, the reduction becomes more pronounced, exceeding $-3.0 \times 10^{11} \text{ W} \cdot \text{m}^{-3}$, as shown in Fig. 11(b)–(f). Furthermore, as the distance from the tungsten electrode tips increases, the two curves gradually converge. However, the magnified views in Fig. 11 reveal that for shallow keyhole depths, a considerable difference reappears between the two curves near the keyhole bottom, as illustrated in Fig. 11(a)–(c). This difference decreases with increasing keyhole depth. The primary reason is that the large temperature difference between the

arc plasma and the BM generates a steep temperature gradient in this region, thereby increasing the electron transport enthalpy.

For deeper keyholes, as shown in the magnified views of Fig. 11(d)–(f), the two curves almost coincide. Although the temperature gradient near the keyhole bottom remains large, the current predominantly exits through the upper sidewall of the keyhole (Fig. 20 and Fig. 21), leading to a lower current density in this region (Fig. 19) and consequently a significant reduction in electron transport enthalpy, resulting in the convergence of the two curves.

Fig. 12 presents the thermal radiation loss profiles along the symmetry axis at different keyhole depths, with and without considering the effect of metal vapour. It is evident from Fig. 12 that the inclusion of metal vapour results in a substantial reduction in thermal radiation loss. When the keyhole depth is 0 mm (Fig. 12(a)), the maximum value decreases from $-1.14 \times 10^{11} \text{ W} \cdot \text{m}^{-3}$ to $-4.55 \times 10^{10} \text{ W} \cdot \text{m}^{-3}$, corresponding to a reduction of $-6.85 \times 10^{10} \text{ W} \cdot \text{m}^{-3}$, i.e., more than a 50% decrease. At other keyhole depths, the magnitude of reduction remains generally consistent.



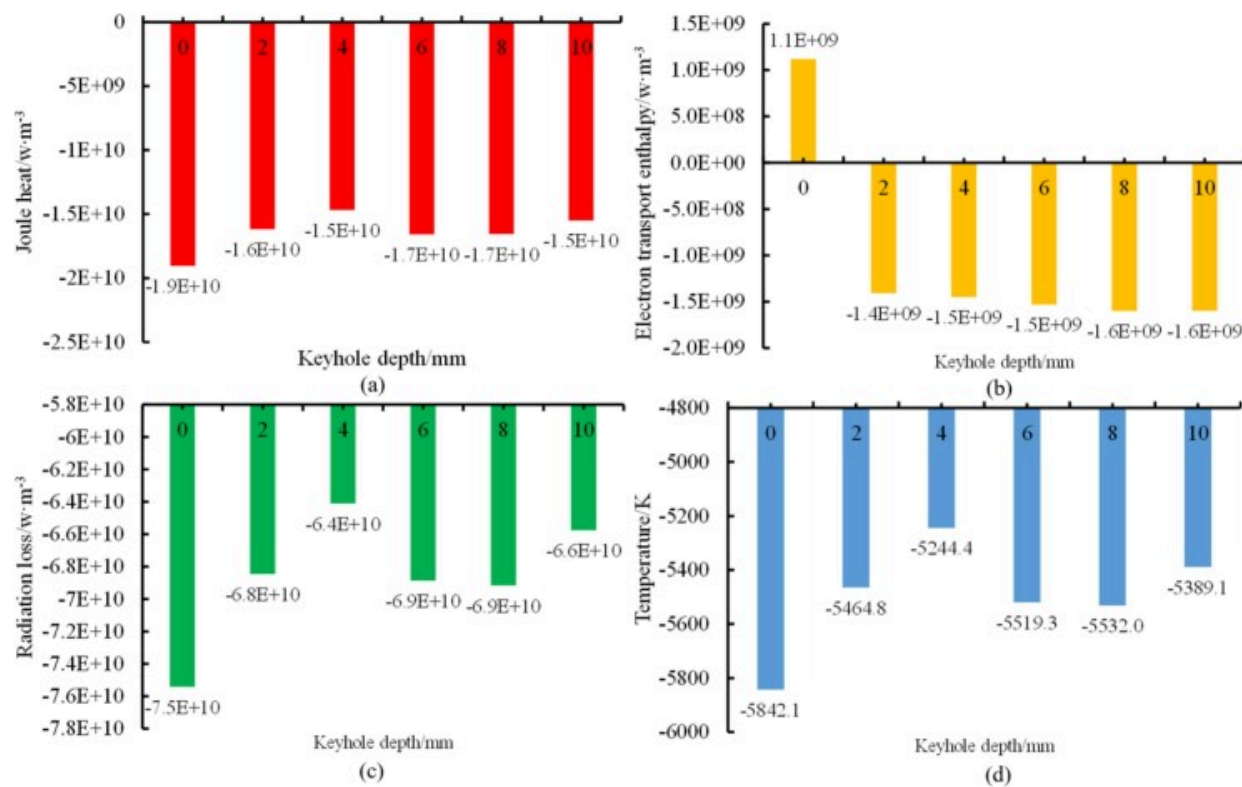
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Fig. 12. Radiation loss along the symmetry axis.

This reduction may be attributed to the following mechanism: the presence of metal vapour increases the electrical conductivity of the arc plasma, which in turn reduces the current density and consequently decreases Joule heating, leading to a lower temperature. The increased electrical conductivity also results in a reduced temperature gradient, thereby lowering the electron transport enthalpy. According to Fig. 5(f), metal vapour can enhance the thermal radiation loss of arc plasma. However, in this study, the thermal radiation loss decreases. This is likely because the metal vapour content is relatively low, and its contribution to increasing thermal radiation loss is smaller than the reduction caused by the temperature drop in the arc plasma due to decreased Joule heating. Consequently, the net effect is a decrease in thermal radiation loss.

To further quantify the temperature variation of the arc plasma with and without considering metal vapour, thermal parameters were extracted along the symmetry axis at a position 1.5 mm ($y=8.5$ mm) from the tungsten electrode tip for different keyhole depths. The difference between the values obtained with and without metal vapour was calculated to assess its influence and contribution to temperature variation, as illustrated in Fig. 13.



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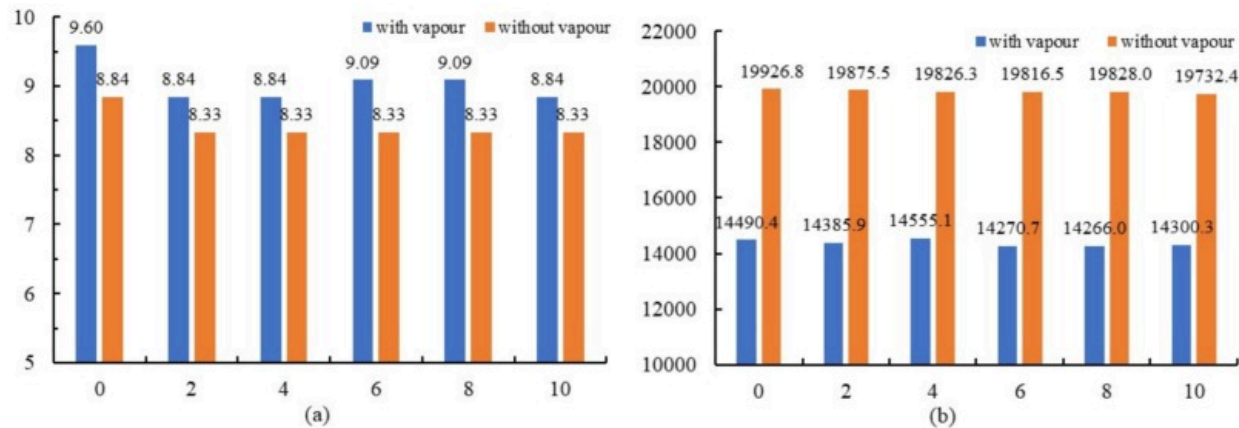
Fig. 13. Variation of energy transport difference before and after considering metal vapour at different keyhole depths (a) Joule heat (b) Electron transport enthalpy (c) Radiation loss (d) Temperature.

As shown in Fig. 13(a), although Joule heating decreases at all keyhole depths when metal vapour is considered, the magnitude of this reduction remains nearly constant at approximately $1.7 \times 10^{10} \text{W} \cdot \text{m}^{-3}$. Joule heating serves as the primary heat source of the arc plasma. Specifically, at a keyhole depth of 0mm, the electron transport enthalpy increases by $1.1 \times 10^9 \text{W} \cdot \text{m}^{-3}$ (Fig. 13(b)), while the thermal radiation loss decreases by $7.5 \times 10^{10} \text{W} \cdot \text{m}^{-3}$ (Fig. 13(c)), representing the largest reduction across all keyhole depths. Nevertheless, the temperature still decreases by 5842.1 K (Fig. 13(d)), the greatest temperature drop observed. This behavior is likely due to the dominant influence of Joule heating on arc plasma temperature: at 0mm keyhole depth, Joule heating exhibits its largest reduction ($1.9 \times 10^{10} \text{W} \cdot \text{m}^{-3}$), which outweighs the effects of increased electron transport enthalpy and reduced thermal radiation loss, leading to a net temperature decrease.

With increasing keyhole depth, the electron transport enthalpy shifts from an increase to a decrease, with a reduction of approximately $1.5 \times 10^9 \text{W} \cdot \text{m}^{-3}$ (Fig. 13(b)). At the same time, the thermal radiation loss continues to decrease but with a smaller magnitude of about $6.6 \times 10^{10} \text{W} \cdot \text{m}^{-3}$ (Fig. 13(c)), while the reduction in Joule heating also becomes smaller, around $1.7 \times 10^{10} \text{W} \cdot \text{m}^{-3}$ (Fig. 13(a)). Consequently, the magnitude of temperature reduction in the arc plasma decreases to approximately 5400K (Fig. 13(d)).

The comparison of the highest temperature points in the arc plasma before and after considering the metal vapour at different keyhole depths is shown in Fig. 14. Fig. 14(a) represents the comparison of the highest temperature location points, while Fig. 14(b) presents a comparative analysis of peak temperature values. It is revealed from Fig. 14(a) that the highest temperature is not located at the tungsten electrode tips. According to Eq. (22), this is due to the fact that the thermal energy of the arc plasma consists of three components: Joule heat, electron transport enthalpy, and radiative heat loss, which together contribute to the temperature distribution. At the tungsten electrode tips, although the current density is high — as shown in Fig. 19 — the electron transport enthalpy is closely related to the temperature gradient. As illustrated in Fig. 9, the temperature increases progressively from the tungsten electrode outward, indicating a

negative temperature gradient and thus a negative electron transport enthalpy, as shown in Fig. 11. Therefore, the electron transport enthalpy decreases gradually as it approaches the tips of the tungsten electrode. In this region, the high current density results in significant Joule heating. However, radiative heat loss and electron transport enthalpy are negative, and their values are larger, it is sufficient to offset the effect of Joule heating, leading to a lower temperature.



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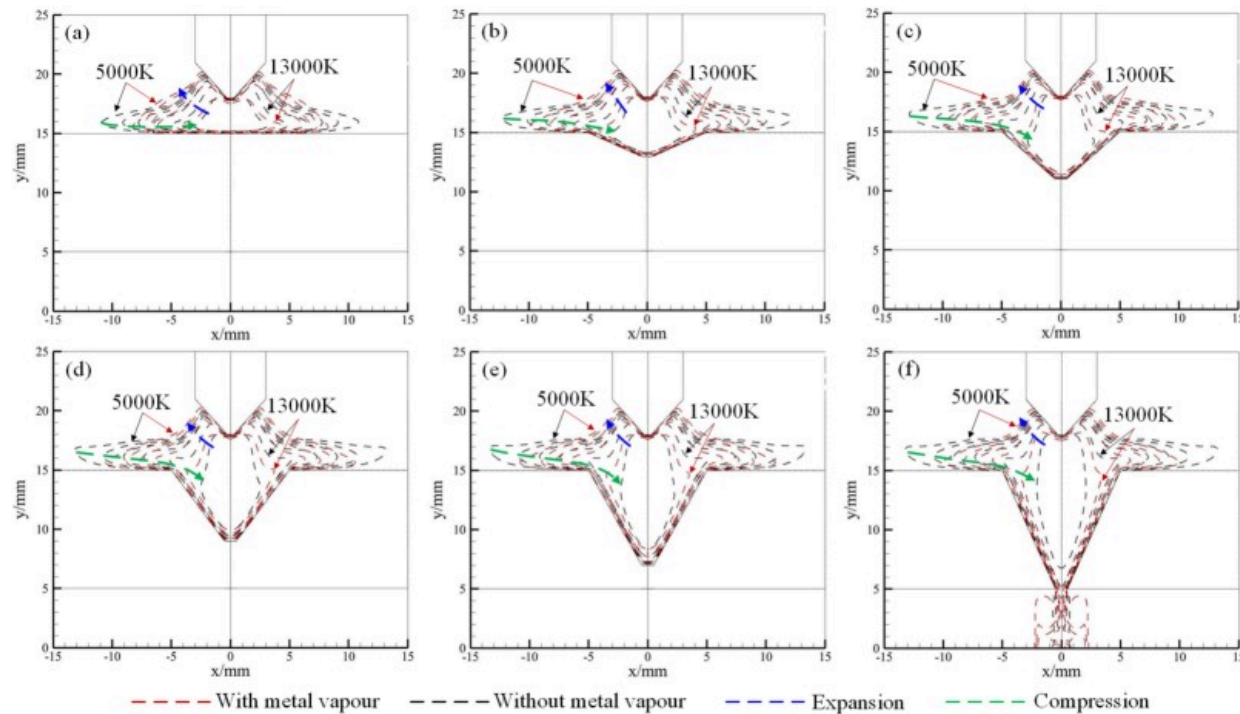
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Fig. 14. Comparison of the highest temperature points.

Furthermore, it can be observed from Fig. 14 that at different keyhole depths, the distance from the highest temperature point to the tungsten electrode tip increases when metal vapour is considered, compared to when it is not. This is because as the distance from the tungsten electrode tip increases, the current density gradually decreases, resulting in a corresponding reduction in Joule heating. Meanwhile, radiative heat loss remains negative but reduce gradually, and electron transport enthalpy gradually decrease to 0 from negative. Therefore, although the Joule heat decreases as the distance from the tungsten electrode increases, the radiation heat loss and electron transport enthalpy decrease, resulting in a gradual increase in temperature, leads to a shift in the location of the peak temperature, especially when metal vapour is considered.

From Fig. 14, it can also be observed that when the keyhole depth is 0mm, the highest temperature point is furthest from the tungsten electrode tip, at $z=8.84\text{mm}$ (without metal vapour) and $z=9.60\text{mm}$ (with metal vapour), while at other keyhole depths, the position of the highest temperature point remains essentially the same, at $z=8.33\text{mm}$ (without metal vapour) and $z=8.84\text{mm}$ or 9.09mm (with metal vapour). The explanation for this phenomenon is as follows: when the keyhole depth is 0mm, the axial electromagnetic force is relatively weak, as shown in Fig. 34. This results in a smaller axial acceleration and the absence of mechanical compression effects from the keyhole. Consequently, the plasma axial velocity is lower, as indicated in Fig. 30, as a result, the peak temperature location appears higher compared to cases with greater keyhole depths.

To further investigate the effect of metal vapour on the arc plasma temperature distribution, we obtained the temperature contour lines of the arc plasma at various keyhole depths for comparison, as presented in Fig. 15. Keyhole deepening induces morphological changes in arc plasma isotherms, featuring basal depression with simultaneous axial extension and radial compression. This is primarily caused by the mechanical compression from the keyhole formation on the arc plasma. However, the formation of the keyhole also helps the arc plasma to penetrate deeper, thereby aiding in the heating of the BM on both sides of the keyhole, which improves thermal energy utilization efficiency and, consequently, enhances welding efficiency.



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Fig. 15. Comparison of arc plasma temperature contour lines.

Furthermore, as illustrated in Fig. 15, the presence of metal vapour leads to noticeable deformation of the thermal field. Specifically, the low - temperature isotherms near the tungsten electrode tips expand outward, whereas the high - temperature core region and the isotherms near the BM contract inward, as indicated by the green and blue arrows in Fig. 15. This phenomenon is closely related to the electrical conductivity of the arc plasma.

As shown in Fig. 5(d), when the temperature is below 5000K, the electrical conductivity of the arc plasma remains nearly the same regardless of the presence of metal vapour. However, within the range of 5000K to 15,000K, the addition of metal vapour

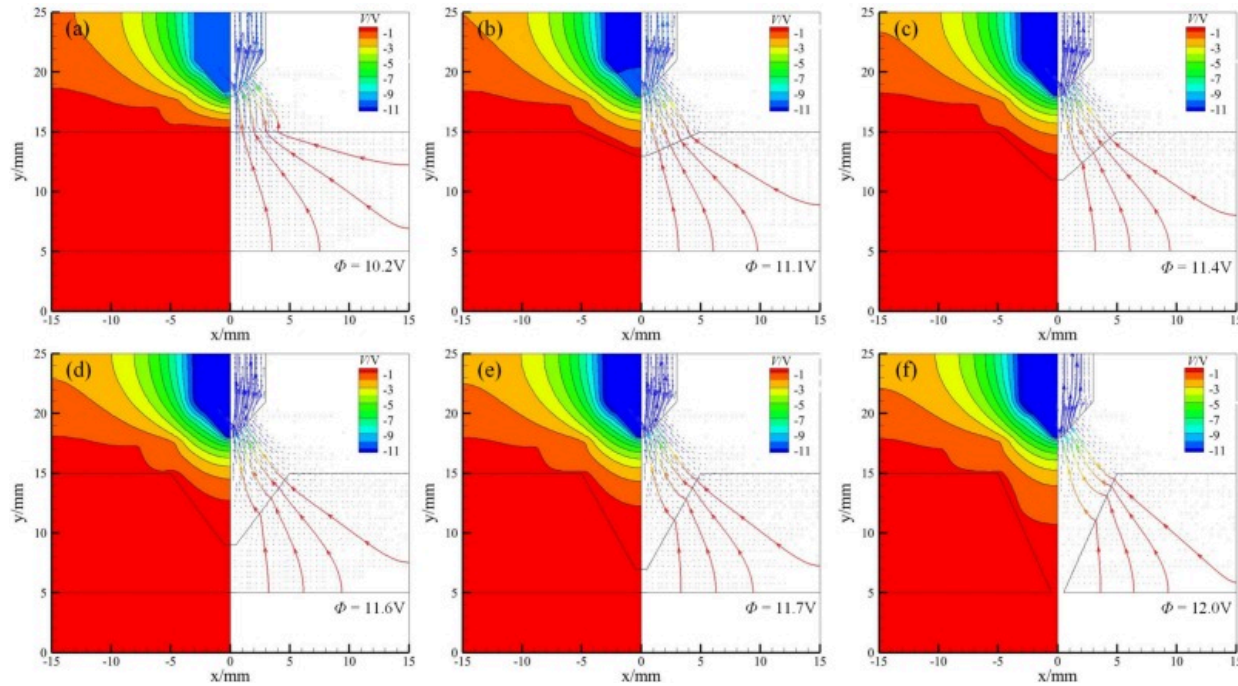
significantly increases the electrical conductivity. As the temperature increases further, the effect of metal vapour becomes more nuanced: at low metal vapour concentrations, the electrical conductivity continues to increase with temperature, whereas at high vapour concentrations, it starts to decrease. In the present study, [Fig. 8](#) demonstrates that the maximum arc plasma temperature remains below 15,000K. Therefore, within this temperature range, the electrical conductivity is positively correlated with temperature.

In the high - temperature core region, the addition of metal vapour enhances electrical conductivity, i.e., improves electrical transport properties. According to Eq. (22), higher electrical conductivity leads to reduced Joule heating. Consequently, the reduced heat input causes the isotherms in this region to contract inward.

In the region near the tungsten electrode, [Fig. 7](#) and [Fig. 8](#) indicate that the temperature gradient is steep, and [Fig. 9](#) shows that the local temperature is below 5000K. Although the change in electrical conductivity is relatively small in this range, the presence of metal vapour still causes a slight increase in electrical conductivity. This broadens the conductive channel, thereby leading to the outward expansion of the low - temperature isotherms.

3.2. Effect of Fe metal vapour on potential and current density field

As depicted in [Fig. 16](#), the potential and current trajectory distribution at different keyhole depths at the 500ms are depicted, without considering the effect of metal vapour. The potential increases with the keyhole depth, rising from 10.2V at a depth of 0mm to 12.0V at a depth of 10mm, a total increase of 1.8V, or 17.6%. This is due to the fact that as the keyhole is formed, the length of the arc plasma gradually increases, thus lengthening the conductive path and causing the potential to increase.



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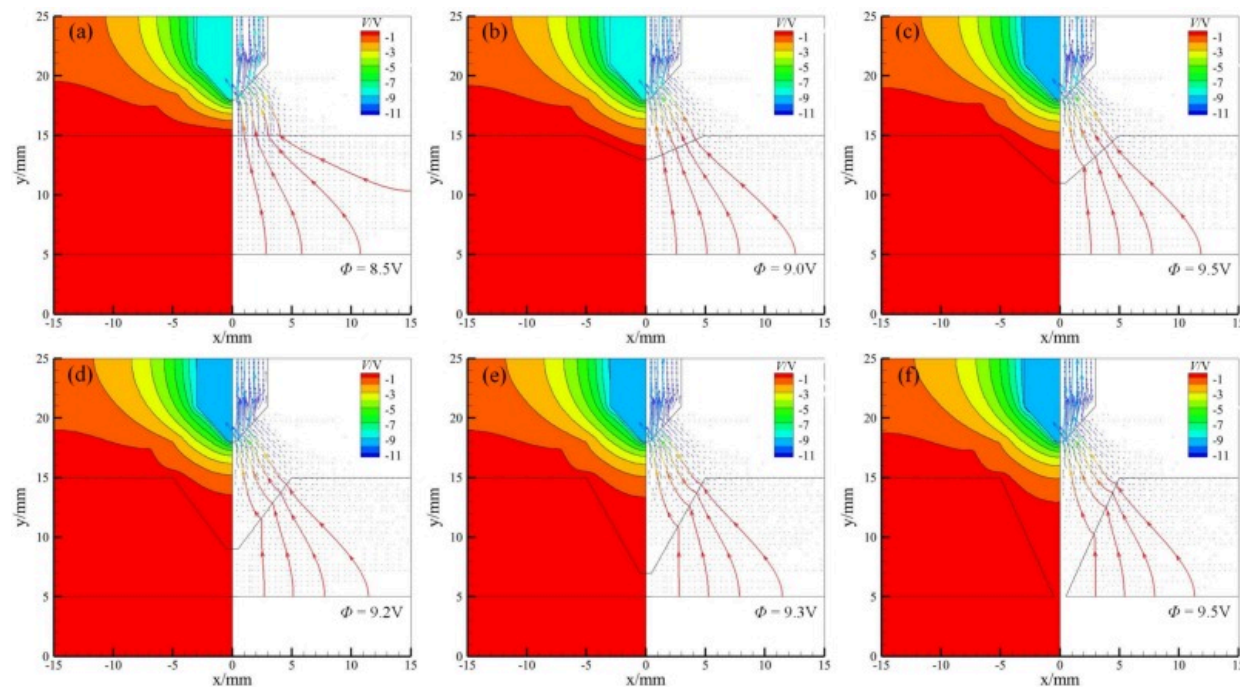
Fig. 16. Arc potential field (without metal vapour).

From Fig. 16, it can also be observed that in the area near the BM, the potential is essentially zero and changes very little. However, as the distance from the BM increases and the proximity to the tungsten electrode grows, the potential variations become more pronounced. This phenomenon arises from the upward current path from BM to tungsten electrode, coupled with decreased conductive cross - section near the cathode. Consequently, the current density becomes increased, and according to Eqs. (8), (9), the potential change becomes more pronounced.

From Fig. 16, it is also observed that the current lines converge toward the center along the keyhole's side walls, rather than passing through the keyhole bottom. This is mainly because the side walls and bottom of the BM are at zero potential. In accordance with the

minimum voltage principle, as the arc plasma penetrates further into the keyhole, it automatically seeks the shortest path, minimizing the conductive path. Therefore, it first contacts the side walls located at the upper part of the keyhole, which leads to the observed pattern in Fig. 16.

As depicted in Fig. 17, the potential and current trajectory distribution at different keyhole depths at the 500ms are depicted, considering the effect of metal vapour. From Fig. 17, it can be observed that the potential is generally lower compared to when metal vapour is not considered. This phenomenon principally stems from metal vapour's ability to elevate the arc's medium - temperature range (5000–15,000K) electrical conductivity, leading to potential reduction.



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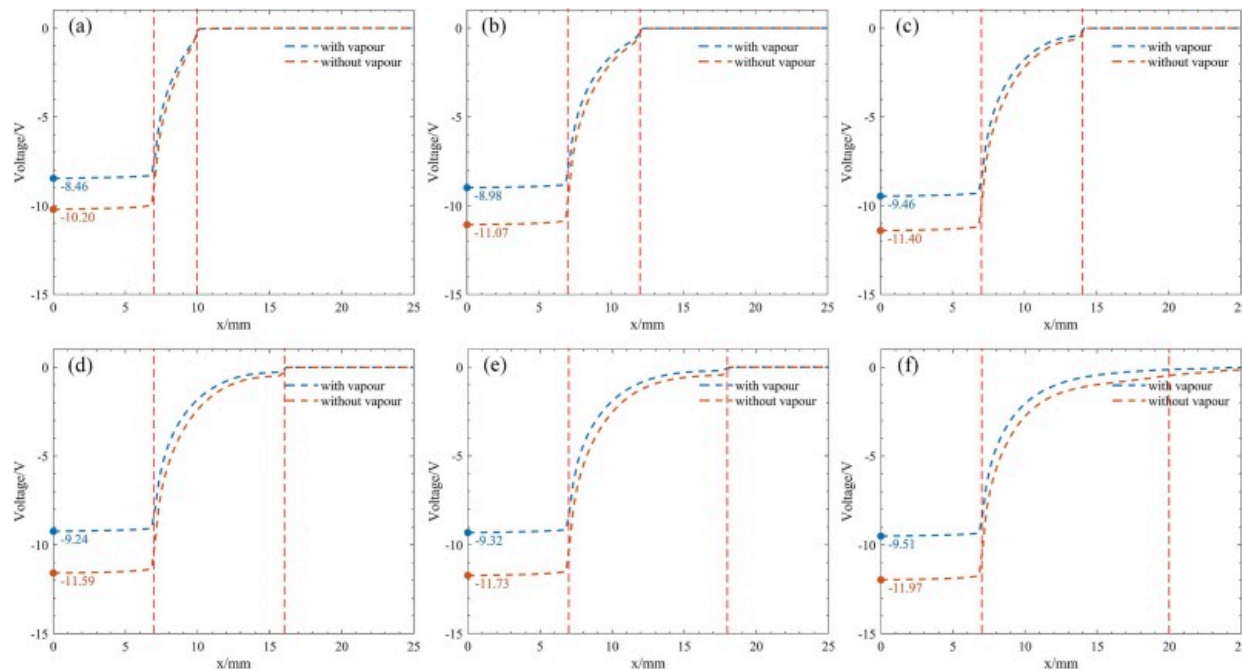
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Fig. 17. Arc potential field (with metal vapour).

Additionally, the potential also changes with variations in keyhole depth. When the keyhole depth is 0mm, the potential is only 8.5V, as shown in Fig. 17(a), while keyhole depth reaches 10mm, the potential increases to 9.5V, as shown in Fig. 17(f), representing a 1V increase, or 11.8%. This increase is lower than the one observed when metal vapour is not considered.

As observed in Fig. 17, potential variation displays oscillatory characteristics with increasing keyhole depth, deviating from a strictly rising pattern. This could be associated with the impact of metal vapour, although the overall variations in potential and current trajectory follow the same pattern as when metal vapour is not considered.

As depicted in Fig. 18, the potential variation curves along the symmetry axis at different keyhole depths at the 500ms, before and after considering the effect of metal vapour, are presented. As previously mentioned, after considering metal vapour, the potential decreases, leading to a significant difference between the two curves.



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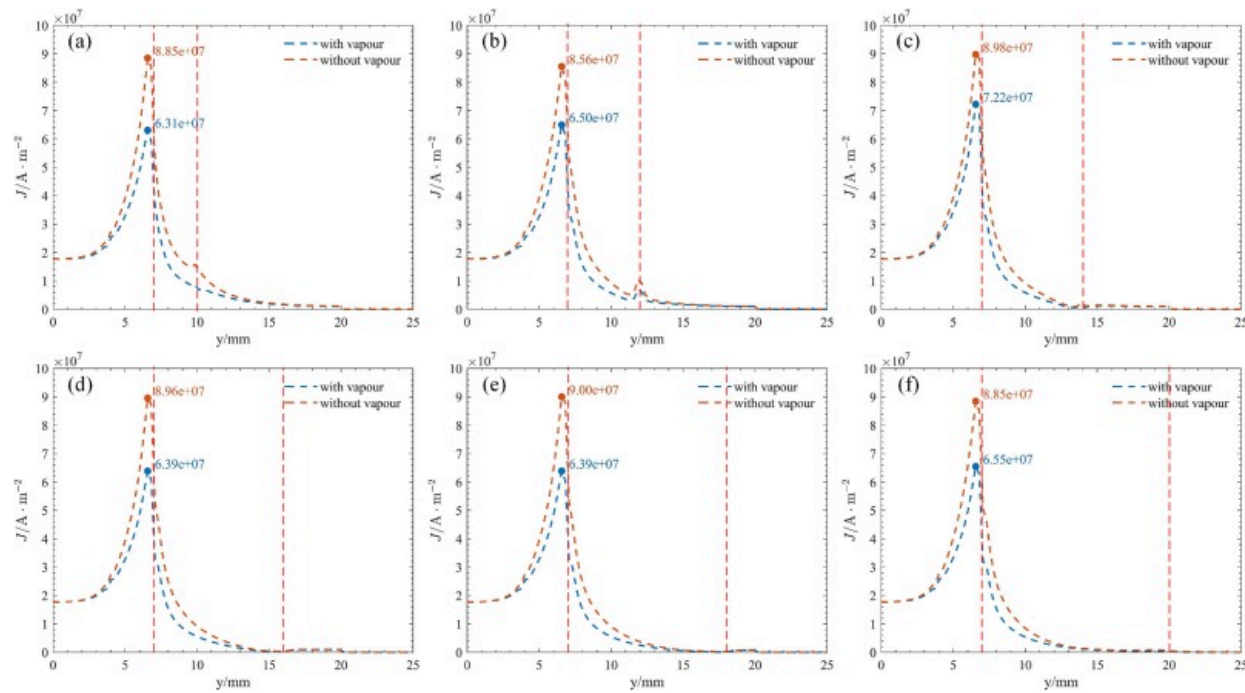
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Fig. 18. Potential curves along the symmetry axis.

However, upon closer inspection, it can be observed that in the arc plasma column region, the potential curves are nearly identical, with only a small difference, while adjacent to the tungsten electrode the discrepancy between the curves increases significantly. The observed phenomenon results from the following contributing factors: As depicted in [Fig. 7](#), [Fig. 8](#) and [Fig. 9](#), the temperature in the arc plasma column region ranges between 14,000K and 20,000K, where the electrical conductivity of the arc plasma does not differ significantly. Consequently, the potential curves nearly coincide in this region.

In contrast, the arc plasma temperature plummets sharply near the tungsten electrode ([Fig. 9](#)), leading to a drastic change in its electrical conductivity, as depicted in [Fig. 5\(d\)](#). The combined effect of these factors causes the discrepancy between the two potential curves to increase significantly in this region. Once entering the tungsten electrode region, where temperature variations are minimal, the potential curves tend to become parallel. Similarly, in the BM domain, where the electrical conductivity is set to a constant, the potential curves overlap.

As depicted in [Fig. 19](#), the current density curves along the symmetry axis at different keyhole depths at the 500ms, with and without considering the effect of metal vapour, are displayed. As depicted in [Fig. 19](#), the tungsten electrode tip exhibits the highest current density concentration, with rapid attenuation observed in both sides, almost $9.0 \times 10^7 \text{ A} \cdot \text{m}^{-2}$ and $6.5 \times 10^7 \text{ A} \cdot \text{m}^{-2}$, respectively.



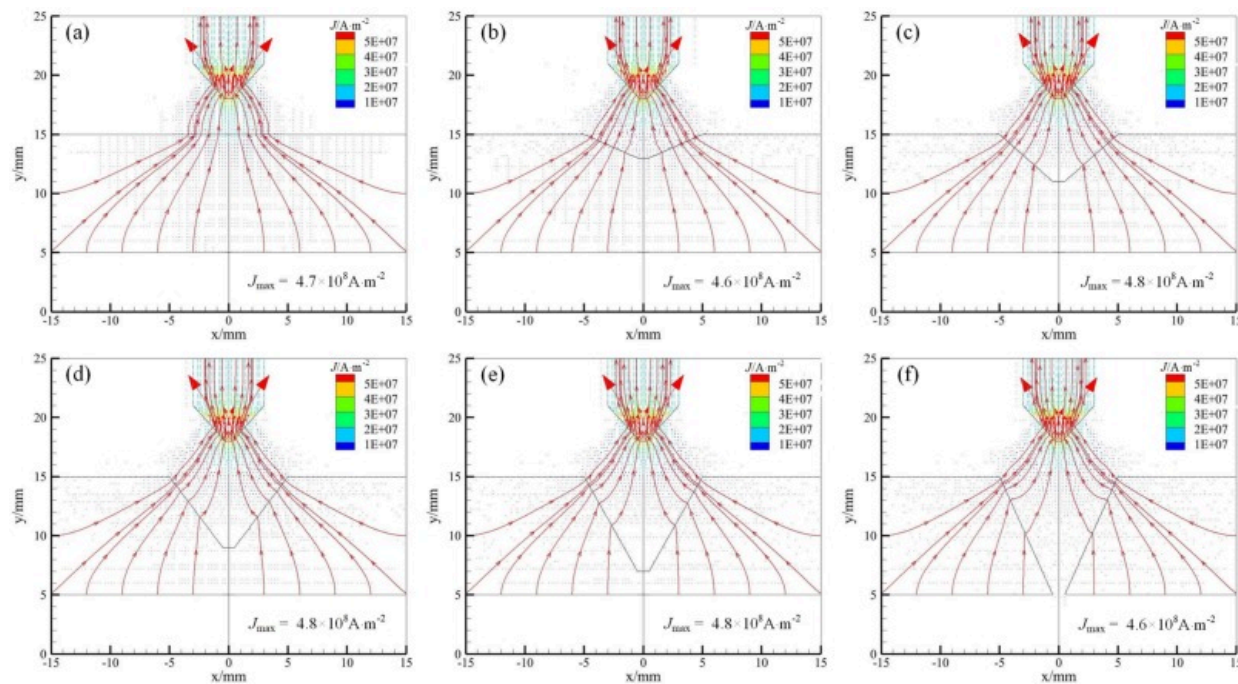
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Fig. 19. Current density curves along the symmetry axis.

According to Eq. (22), this sharp decrease in current density leads to a rapid reduction in Joule heating, causing a corresponding drop in temperature and, consequently, changes in electrical conductivity. As a result, the potential difference in this region increases rapidly. Furthermore, from Fig. 19, the results demonstrate that without accounting for metal vapour, the current density at this point is approximately $9.0 \times 10^7 \text{ A} \cdot \text{m}^{-2}$, which is higher compared to when metal vapour is considered ($6.5 \times 10^7 \text{ A} \cdot \text{m}^{-2}$). Therefore, the current density slope on both sides decreases more sharply, causing the potential to change more rapidly and thus be higher. In the arc plasma column region, where the current density remains relatively stable, the potential curves exhibit minimal differences, as depicted in Fig. 18.

To gain a deeper understanding of the current density variation, a distinct analysis is carried out regarding the distribution of current density vectors. As depicted in Fig. 20, when the influence of metal vapour is neglected, the current density vector diagram at the 500ms for different keyhole depths is presented. The maximum current density is approximately $4.7 \times 10^8 \text{ A} \cdot \text{m}^{-2}$, located in the tungsten electrode tip region, as indicated by the red vector arrows, which is significantly higher than that in other regions.



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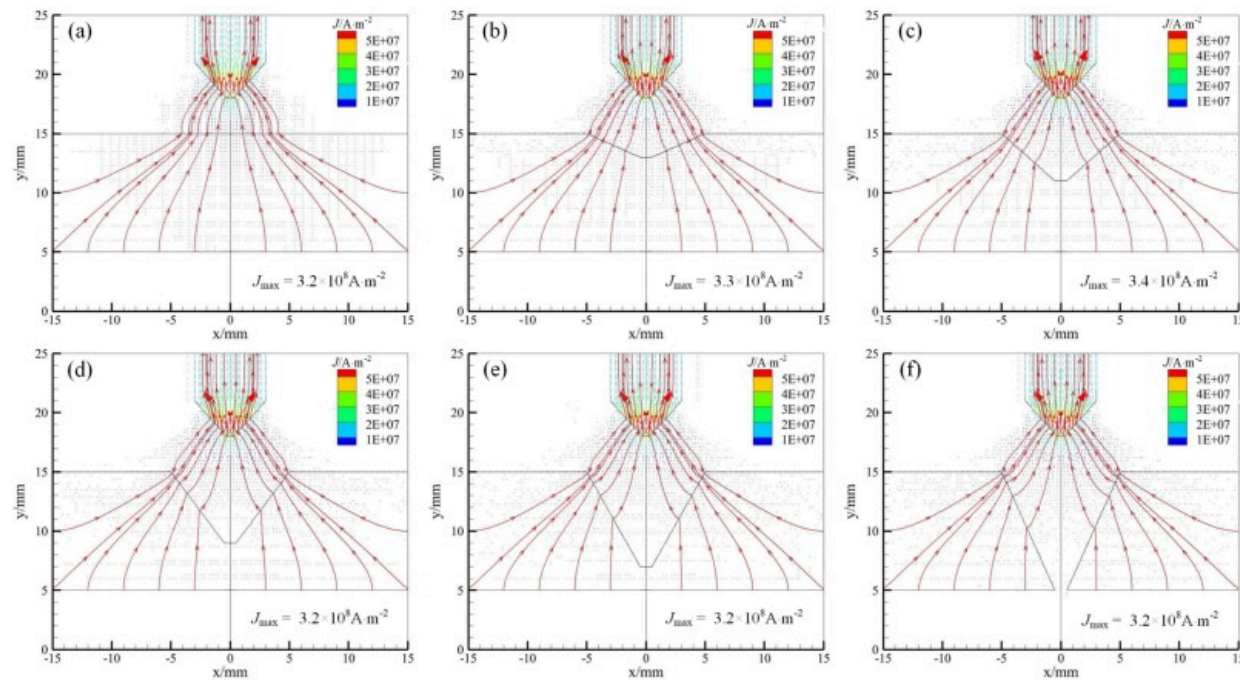
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Fig. 20. Current density vector (without metal vapour).

Fig. 20 clearly illustrates that the current preferentially flows along the upper sidewalls of the keyhole, especially at the corners of the keyhole entrance, while they are less dense at the keyhole bottom. The current density distribution significantly influences the heat energy transfer from the arc plasma to the BM. Due to this trajectory pattern, the upper side walls of the keyhole receive more Joule heat, whereas the bottom receives relatively

less. As a result, the high - temperature area of the BM takes on an open “V” shape, as depicted in Fig. 7(a).

As depicted in Fig. 21, the current density vector diagram at different keyhole depths at the 500ms, considering the effect of metal vapour, is presented. The maximum current density is approximately $3.3 \times 10^8 \text{ A} \cdot \text{m}^{-2}$, which is lower compared to when metal vapour is not considered. However, since the current boundary conditions used in the simulation remain unchanged, this effect stems from metal vapour - induced improvement in the arc's medium - temperature range (5000–15,000K) electrical conductive properties. In the tungsten electrode tip region, where the temperature is relatively lower, the enhanced electrical conductivity allows the current to flow through a larger area. Consequently, under the same current boundary conditions, the current density decreases. Other variation trends remain consistent with those observed in the absence of metal vapour.

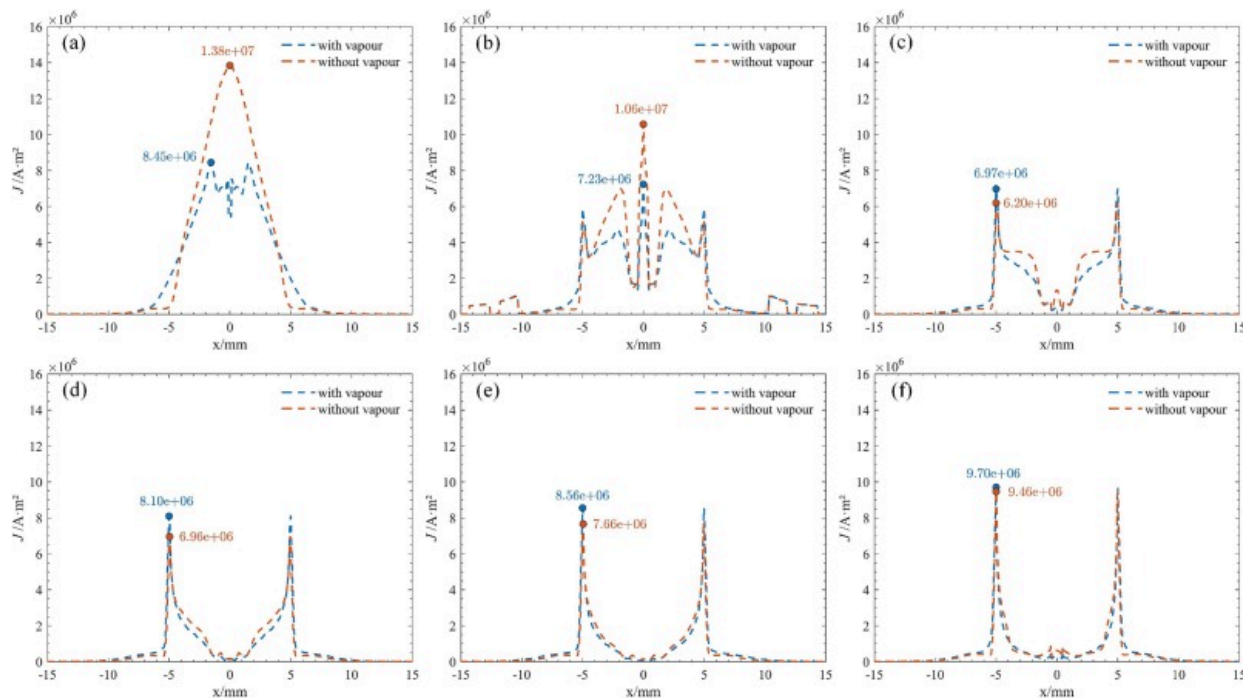


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Fig. 21. Current density vector (with metal vapour).

As shown in Fig. 22, the distribution of current density curves on the upper surface of the BM at different keyhole depths, comparative results with and without metal vapour consideration are presented. When the keyhole depth is 0mm, the current density distribution exhibits the most significant difference between the two cases. Without considering metal vapour, the distribution is unimodal, with a maximum value of $1.38 \times 10^7 \text{ A}\cdot\text{m}^{-2}$. However, when metal vapour is considered, the distribution transforms into a bimodal pattern, and the maximum value significantly decreases to $8.45 \times 10^6 \text{ A}\cdot\text{m}^{-2}$, as depicted in Fig. 22(a). This occurs because metal vapour enhances the medium - temperature range (5000–15,000K) electrical conductivity of the arc plasma while decreasing its high - temperature electrical conductivity. Consequently, after accounting for metal vapour, the current density increases in the medium - temperature range (5000–15,000K) areas while decreasing under the core high - temperature area.



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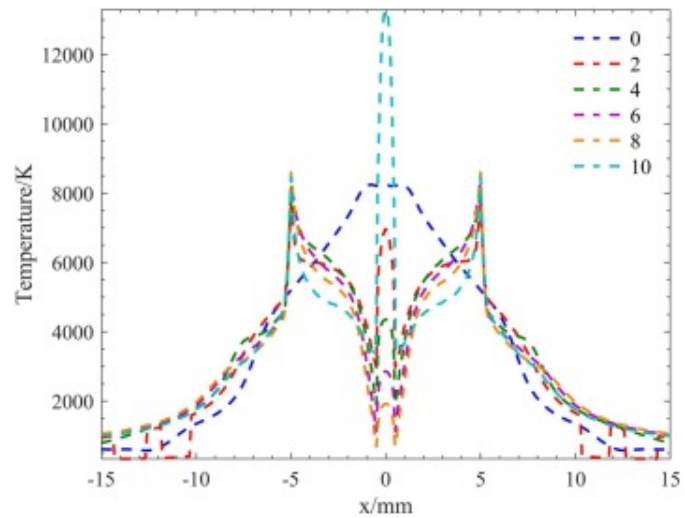
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Fig. 22. Current density curve on the upper surface of the keyhole.

As the keyhole depth grows to 2mm, the current density maximum value decreases sharply, from $1.06 \times 10^7 \text{ A} \cdot \text{m}^{-2}$ without considering metal vapour to $7.23 \times 10^6 \text{ A} \cdot \text{m}^{-2}$ with metal vapour, as depicted in [Fig. 22\(b\)](#). At this stage, after considering metal vapour, the current density in the core internal region remains lower.

When the keyhole depth increases to 4mm, both current density distributions exhibit a bimodal pattern, with the maximum values occurring at the two peaks. The difference between the two cases decreases. At this stage, the current density maximum value is higher when considering the effect of metal vapour. However, in the high - temperature region of the internal core, the current density remains elevated in the absence of metal vapour, as depicted in [Fig. 22\(c\)](#). With increasing keyhole depth, the current density distribution curves for both cases nearly overlap, as depicted in [Fig. 22\(d\)–\(f\)](#), while still maintaining a bimodal distribution.

Further observations indicate that, at this stage, the maximum current density with metal vapour is higher, and maximum current density accumulation occurs at the keyhole aperture's edge regions, significantly exceeding that in other areas. This suggests that the arc plasma generates the most heat at these locations, leading to elevated temperatures that facilitate lateral keyhole expansion. The fundamental cause of this phenomenon can be accounted for as follows: As depicted in [Fig. 23](#), the temperature curve on the upper surface of the keyhole, when considering metal vapour, exhibits higher temperatures at the keyhole corners, although remaining below 15,000K. This indicates that at these locations, the electrical conductivity of the arc plasma is enhanced, resulting in an increase in current density.

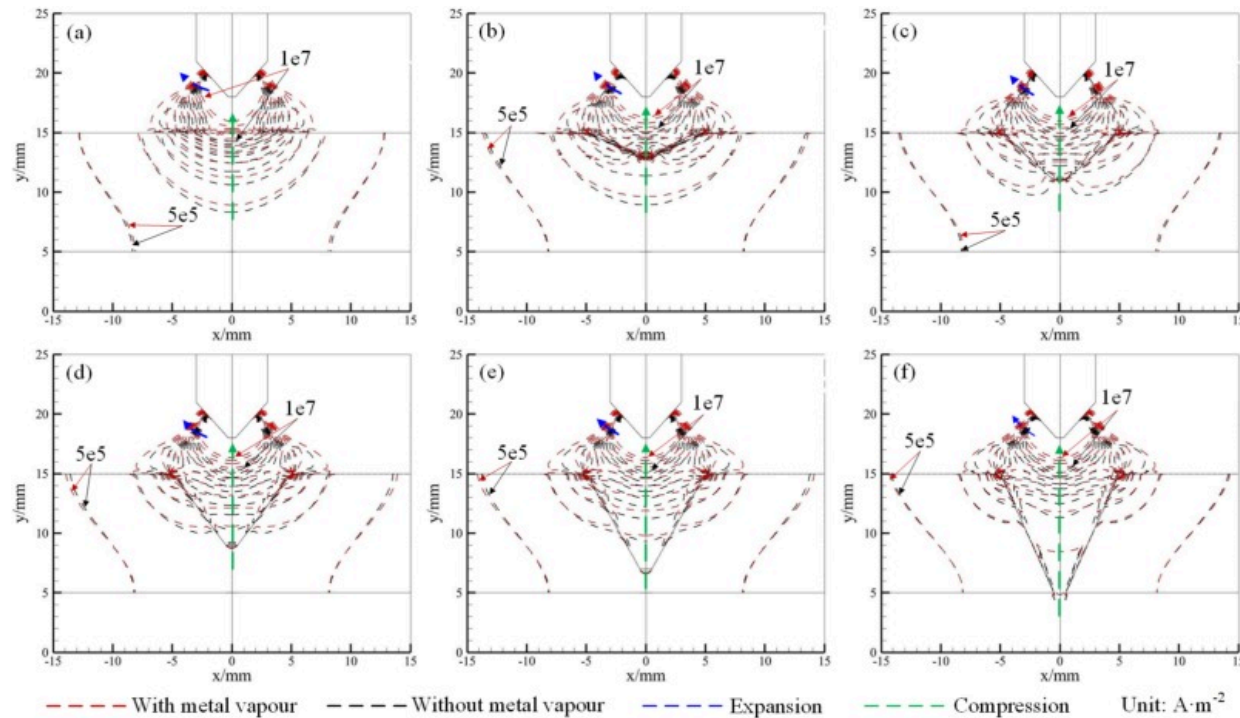


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Fig. 23. Temperature curve on the upper surface of the keyhole (with metal vapour).

As shown in Fig. 24, the current density contour lines for the arc plasma at varying keyhole depths is displayed. From Fig. 24, it is evident that when the impact of metal vapour is considered, the outer current density contour of the arc plasma expands outward, while the inner current density contour contracts inward. This phenomenon is primarily attributed to the relationship between current density and electrical conductivity.



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Fig. 24. Comparison of current density contour lines.

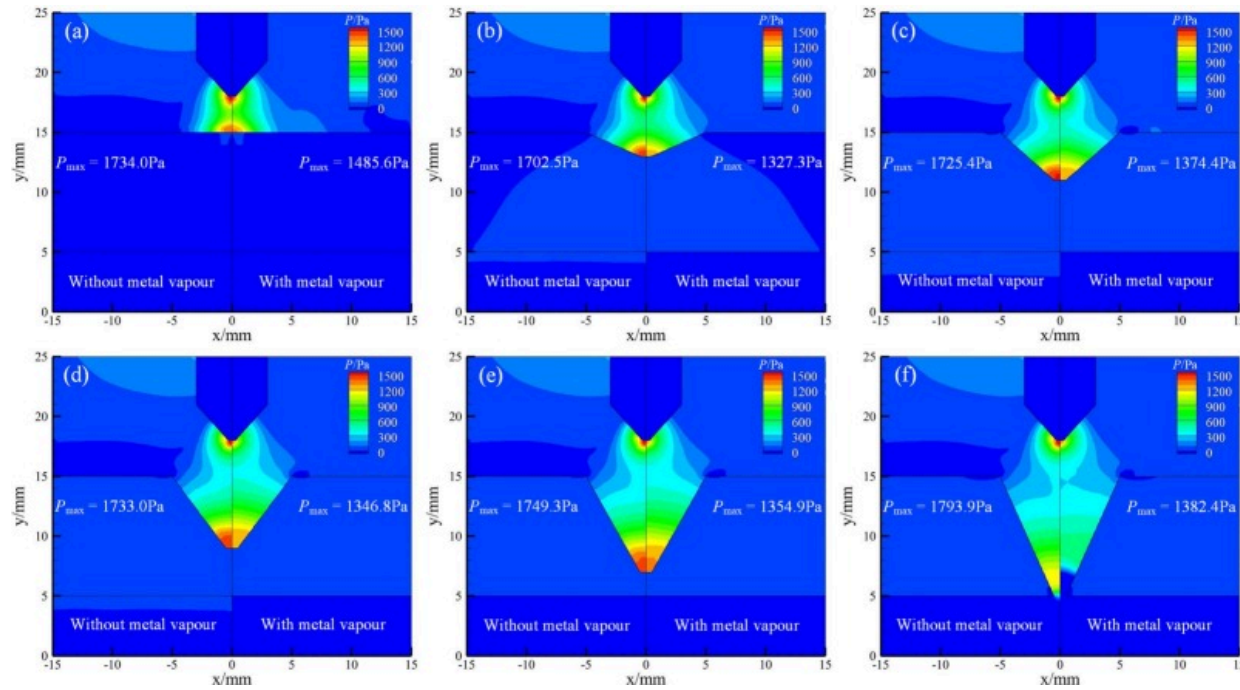
In the outer region, the electrical conductivity is higher when metal vapour is considered compared to when it is not, leading to a broader current density distribution. Conversely, in the internal high - temperature region, where temperatures exceed 14,000K (as shown in Fig. 5(d)), the existence of metal vapour results in a lower electrical conductivity compared to scenarios where metal vapour is not present. Consequently, to achieve the same level of electrical conductivity, higher temperatures are required, leading to the inward contraction of the current density contours.

Furthermore, as shown in Fig. 24, the current density in the BM domain remain nearly unchanged. This is because the electrical conductivity in this region is set as a constant

value of $770,000\text{S}\cdot\text{m}^{-1}$, independent of temperature variations. As a result, the current density contours exhibit minimal change in this region.

3.3. Effect of Fe metal vapour on arc plasma pressure field

As shown in Fig. 25, the arc plasma pressure distribution contour map under different keyhole depths is displayed, both with and without considering the effects of metal vapour. On the left side, the arc plasma pressure distribution contour map is shown without considering the effect of metal vapour, while on the right side, the arc plasma pressure distribution contour map is shown considering the effect of metal vapour. From Fig. 25, it is evident that the high - pressure regions are primarily found at the tungsten electrode tip and the bottom of the keyhole. The elevated arc plasma pressure at the tungsten electrode tip is attributed to the minimal contact area and the high current density present there, which, according to Eqs. (2), (4), generates significant electromagnetic force, as shown in Fig. 32 and Fig. 33, resulting in higher arc plasma pressure, almost 1734.0Pa and 1485.6Pa when keyhole depth is 0mm, respectively. At the bottom of the keyhole, the high pressure is formed due to the blocking effect of the BM, which causes the arc plasma velocity to drastically decrease and change direction. However, when the keyhole depth is 10mm, the arc plasma pressure at the keyhole outlet is greatly reduced because the BM has penetrated. When the metal vapour is taken into account, a negative pressure is even generated there.



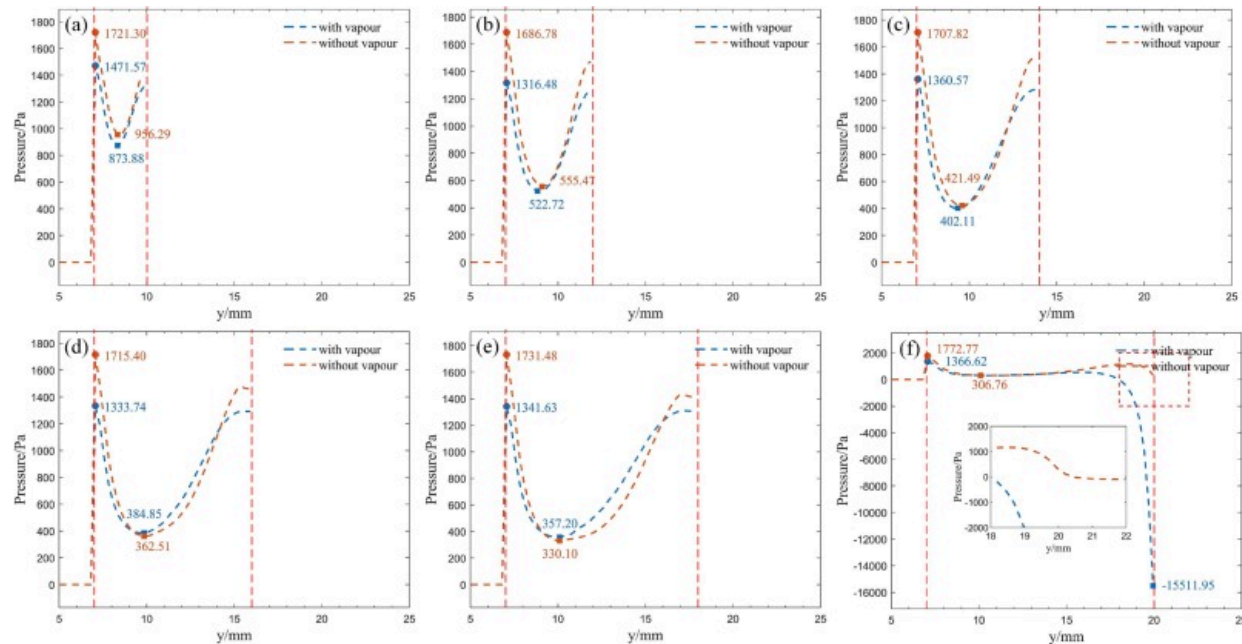
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Fig. 25. Comparison of pressure contour map.

From Fig. 25, it can also be observed that after considering metal vapour effect, the arc plasma pressure significantly decreases. The maximum arc plasma pressure reduces from 1700Pa to approximately 1400Pa under different keyhole depths, representing a decrease of 17.6%. This is primarily due to the increase in the medium - temperature region (5000–15,000K) electrical conductivity of the arc plasma after considering metal vapour. As depicted in Fig. 24, the current density isothermal lines spread out in the vicinity of the tungsten electrode tip. At this point, the current density decreases, as shown in Fig. 20 and Fig. 21, according to Eqs. (2), (4), the electromagnetic force also diminishes, leading to a reduction in arc plasma pressure.

For deeper analysis of arc plasma pressure dynamics, axial arc plasma pressure profiles (with/without metal vapour effects) were extracted at 500ms across varying keyhole depths. as depicted in Fig. 26. From Fig. 26, the highest arc plasma pressure is located at the tungsten electrode tip, reach 1772.77Pa when keyhole depth is 10mm. After considering metal vapour, the highest arc plasma pressure decreases compared to when metal vapour is not considered, only 1366.62Pa when keyhole depth is 10mm. Moreover, at different keyhole depths, the highest arc plasma pressure without considering metal vapour remains almost the same, around 1700Pa. When metal vapour is accounted for, the peak arc plasma pressure reaches 1471.57Pa when keyhole depth is 0mm, as depicted in Fig. 26(a), with increasing keyhole depth, the highest arc plasma pressure decreases to approximately 1310Pa, as depicted in Fig. 26(b)–(f).



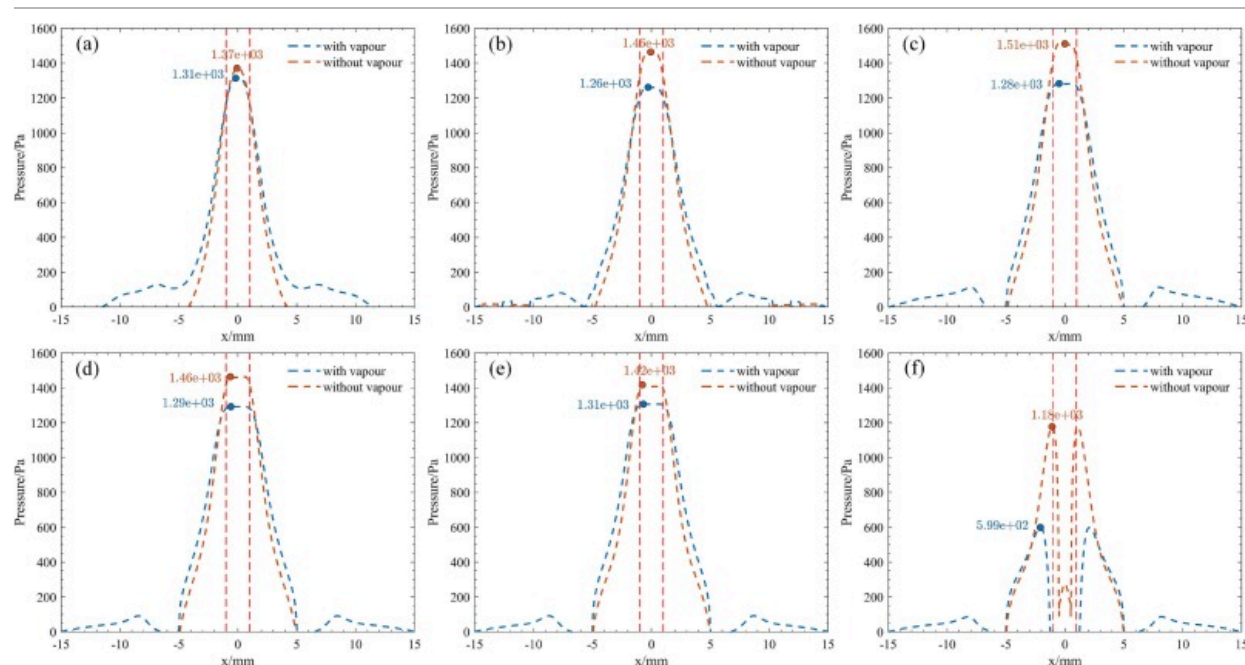
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Fig. 26. Pressure curves along the symmetry axis.

As illustrated in Fig. 26, the arc plasma pressure exhibits a consistent pattern of initial reduction followed by subsequent augmentation at various keyhole depths, both with and without considering metal vapour. Starting from the tungsten electrode tip, the arc plasma pressure gradually decreases, reaching its minimum value in the arc plasma column region, and then gradually increases until it reaches the keyhole bottom. When the keyhole depth attains 10mm, the arc plasma pressure variation follows a normal trend when metal vapour is ignored. However, when the metal vapour effect is accounted for, a significant negative pressure forms at the keyhole outlet, about $-15,511.95\text{Pa}$, and this change in arc plasma pressure will exert a substantial effect on the arc's axial velocity, which will be analyzed in detail later.

Fig. 27 displays comparative arc plasma pressure distribution profiles on the keyhole surface at 500ms, showing cases both with and without metal vapour consideration at varying keyhole depths. From Fig. 27, it can be seen that after considering metal vapour, the arc plasma pressure on the surface of the keyhole decreases. However, upon closer inspection, it can be observed that when the keyhole depth is 0mm, the arc plasma pressure maximum value for both cases is nearly the same, 1370Pa and 1310Pa, respectively, as shown in Fig. 27(a), but the arc plasma pressure distribution becomes more divergent after considering metal vapour. As the keyhole gradually forms, a significant difference between the two cases occurs, with the largest arc plasma pressure differences concentrated near the keyhole tip (the region between the red dashed lines), while the arc plasma pressure in other areas remains almost the same, as depicted in Fig. 27(b)–(e). This phenomenon may be related to the variation in the axial velocity of the arc plasma, which will be explained later.



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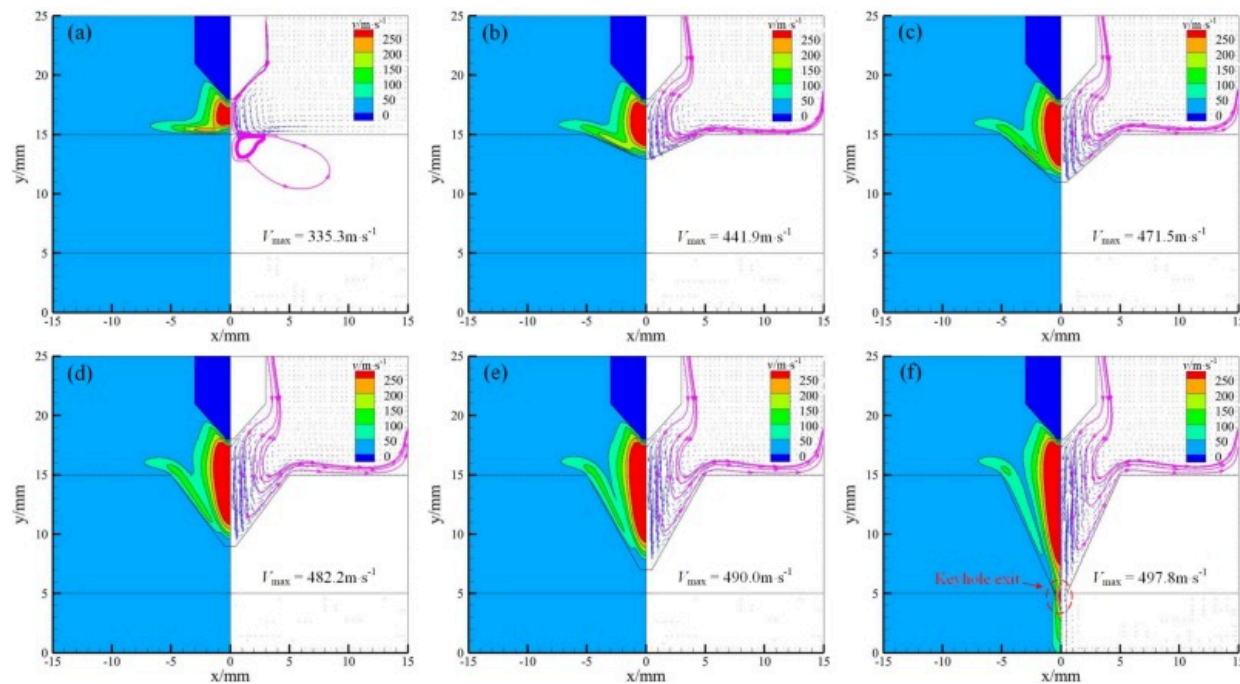
Fig. 27. Pressure curves on the surface of the keyhole.

When the keyhole depth gets to 10mm the maximum difference in arc plasma pressure between the two cases is observed, with the maximum arc plasma pressure in the case without metal vapour being 1180Pa, while the maximum pressure considering metal vapour is only 599Pa, nearly half of the former, as depicted in Fig. 27(f). Notably, after considering metal vapour, a negative pressure zone appears at the keyhole exit.

3.4. Effect of Fe metal vapour on velocity field

Fig. 28 illustrates the arc plasma velocity distribution at 500ms across various keyhole depths under metal vapour exclusion conditions. The left image illustrates the velocity distribution cloud map, whereas the right side presents the velocity vector distribution map. From Fig. 28, it can be seen that as the keyhole depth increases, the velocity of the

arc plasma also continuously increases. The velocity increases from $335.3\text{m}\cdot\text{s}^{-1}$ without a keyhole (0mm), as depicted in Fig. 28(a), to $497.8\text{m}\cdot\text{s}^{-1}$ when the keyhole depth reaches 10mm, as depicted in Fig. 28(f). When the keyhole initially forms (2mm), as depicted in Fig. 28(b), the velocity increase is the largest, reaching $106.6\text{m}\cdot\text{s}^{-1}$, a 31.7% increase. Subsequently, as the keyhole depth further increases, although the velocity also increases, the increase becomes less significant.



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Fig. 28. Velocity distribution contour map (without metal vapour).

The reason for the increase in velocity can be explained as follows: after the keyhole is formed, it exerts a strong confining effect on the arc plasma, causing a large amount of arc plasma to enter the keyhole. Since the internal space is relatively small, according to fluid mechanics theory (the continuity equation Eq. (26) and Bernoulli's equation Eq. (27), identical fluid particles demonstrate velocity magnitudes that are reciprocally related to

conduit cross - section dimensions. When the keyhole is formed, the arc plasma enters the keyhole, forming a mechanical compression effect. Therefore, the area decreases compared to when without keyhole formed, and as depicted in Eq. (26), the velocity increases.

$$A_1 v_1 = A_2 v_2 \quad (26)$$

$$P + \frac{1}{2} \rho v^2 + \rho g h = \text{Constant} \quad (27)$$

where A_1 and A_2 are the cross - sectional areas of the conductors, v_1 and v_2 are the fluid velocities, P represents pressure, v is velocity, and h is the height measured from the liquid surface.

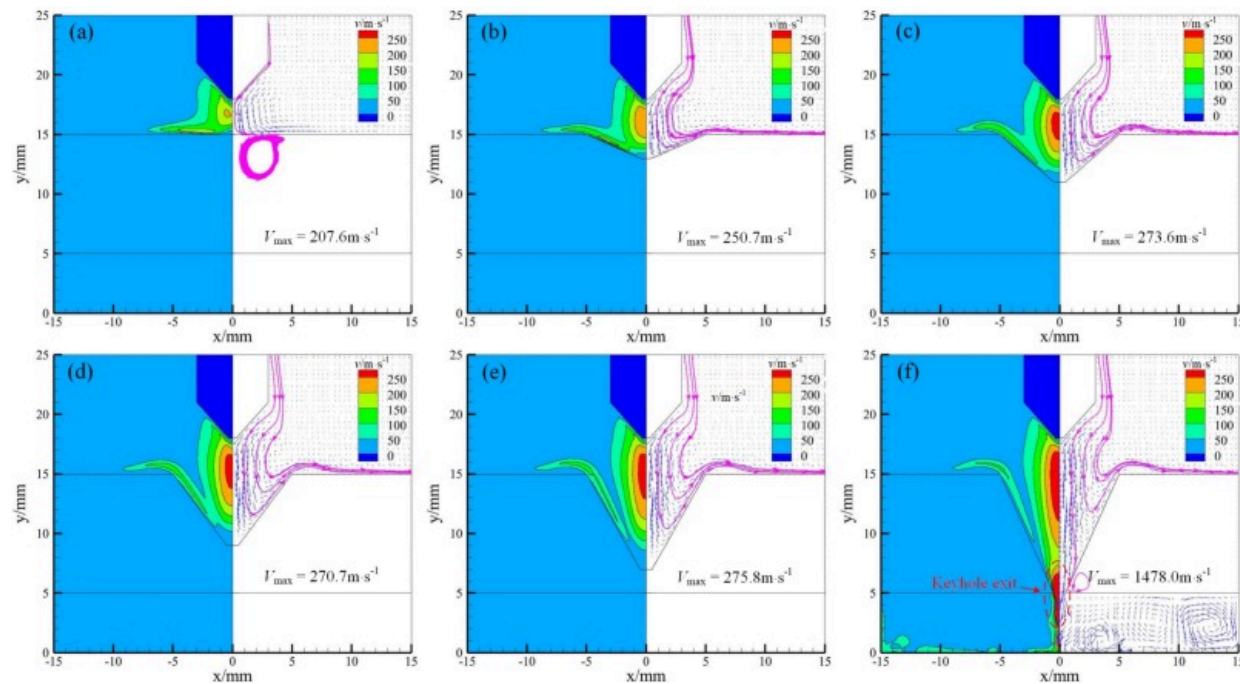
Furthermore, from the velocity vector diagram in Fig. 28, it can be observed that after the Ar shielding gas enters from the top, it first moves along the wall of the tungsten electrode toward the central symmetry axis. Upon reaching the symmetry axis, it then moves vertically downwards and continues along this direction until it reaches the keyhole surface. Owing to the obstruction caused by the keyhole wall, the arc's movement direction is altered, and it proceeds to move along the keyhole surface, moving away from the symmetry axis.

Moreover, from the velocity distribution map in Fig. 28, a significant velocity increase occurs in the arc plasma during a minimal distance at the tungsten electrode, then maintains a high velocity while moving. As the arc plasma nears the keyhole wall, its velocity decreases, and it undergoes a directional shift within a very short distance. Additionally, its velocity gradually decreases radially from the inner region toward the outer region.

When the keyhole depth is 10mm, a small region exists at the keyhole exit where the velocity increases, highlighted by the red ellipse as shown in Fig. 28(f). This may be related to the arc plasma pressure changes on both sides of the keyhole exit. Since the backside surface of the BM is at normal pressure, while the inside of the keyhole is at high

pressure, the arc plasma undergoes an acceleration process when passing through the exit, resulting in an increase in velocity.

Fig. 29 illustrates the arc plasma velocity distribution at 500ms across various keyhole depths when metal vapour is considered. The left image displays the velocity distribution contour map, while the right image presents the velocity vector distribution. It is clearly observed from Fig. 29 that under the same color legend conditions, the velocity distribution contour map appears lighter, indicating that the arc plasma velocity is significantly reduced compared to the case without considering metal vapour.



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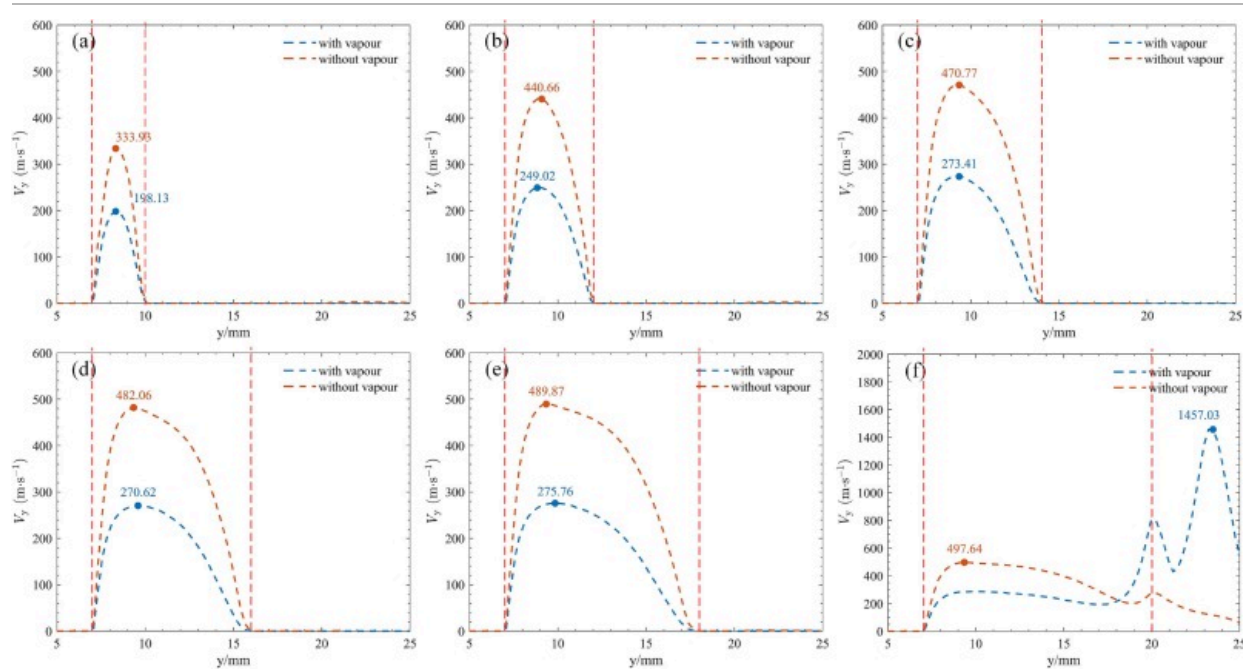
Fig. 29. Velocity distribution contour map (with metal vapour).

When without keyhole formed, the velocity maximum value of arc plasma is only $207.6\text{m}\cdot\text{s}^{-1}$, which is a decrease of $127.7\text{m}\cdot\text{s}^{-1}$ (about a 38% reduction) compared to when

metal vapour is excluded from consideration. Similar to the case without metal vapour, the velocity increases with the keyhole depth. When the keyhole just appears (2mm), the velocity increases by $43.1 \text{ m}\cdot\text{s}^{-1}$, showing a large increase of 20.8%. Subsequently, as the keyhole depth increases further, the velocity increase becomes less noticeable.

Similarly, when the keyhole depth reaches 10mm, there is a sharp velocity increase near the keyhole outlet, as shown by the red elliptical box in [Fig. 29\(f\)](#). This phenomenon is due to the arc plasma pressure difference on both sides of the keyhole outlet, leading to an acceleration of the arc plasma. Further investigation reveals that the maximum velocity increases sharply to $1478.0 \text{ m}\cdot\text{s}^{-1}$, which is approximately three times higher than when metal vapour is not considered. This is primarily due to a greater arc plasma pressure difference at the keyhole outlet with the consideration of metal vapour, as depicted in [Fig. 25\(f\)](#), resulting in a larger velocity increase. Furthermore, this occurrence could potentially be attributed to electromagnetic force effects, as will be elaborated subsequently through analysis of the electromagnetic force distribution.

To further investigate the axial velocity variation of the arc plasma with and without considering the metal vapour effect, the axial velocity (v_y) distribution along the symmetry axis of the arc plasma at different keyhole depths at 500ms was extracted, as shown in [Fig. 30](#). It is evident from [Fig. 30](#) that the axial velocity of the arc plasma rises with the keyhole depth. After considering the metal vapour, the arc's axial velocity is significantly lower than when metal vapour is not considered, and this aligns well with the contour maps of velocity distribution shown in [Fig. 28](#) and [Fig. 29](#).



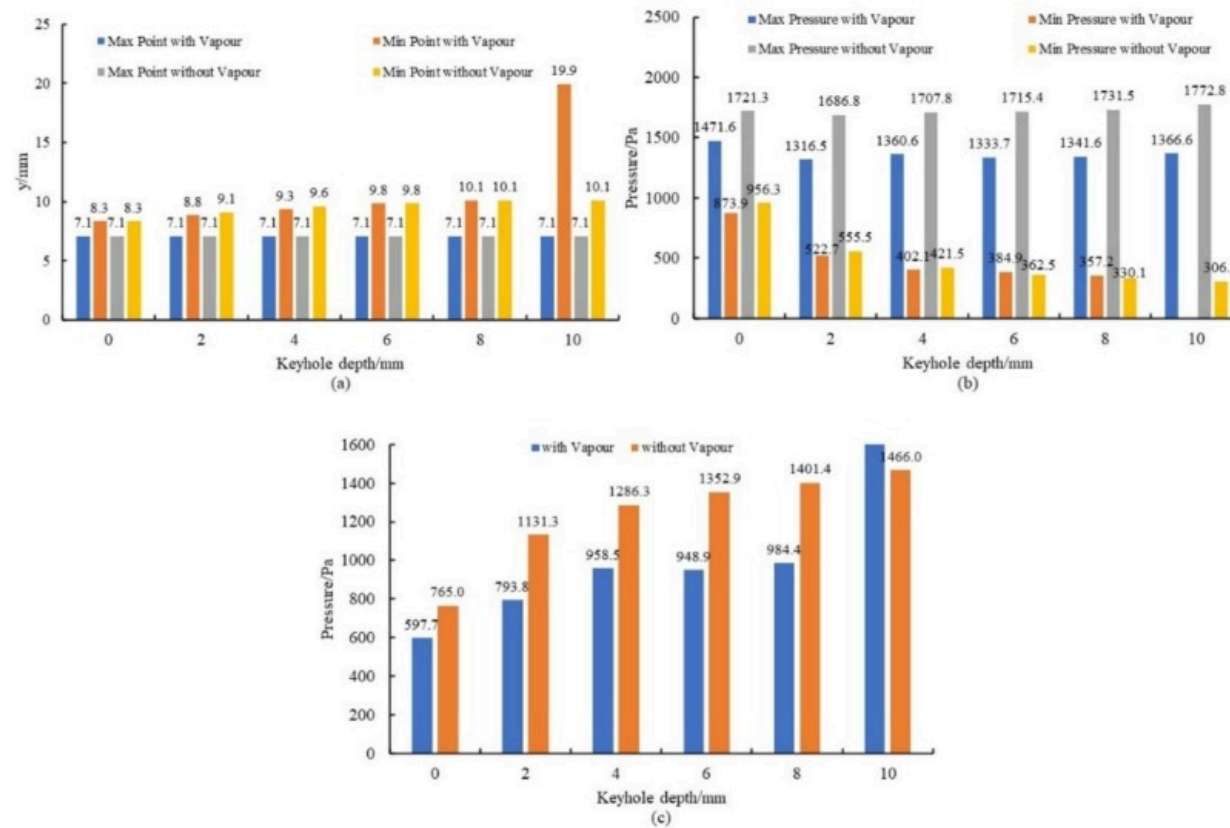
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Fig. 30. Axial velocity profiles along the symmetry axis.

Furthermore, in the arc plasma region (between the red dashed lines), starting from the tungsten electrode ($y=7\text{ mm}$), the arc's axial velocity experiences rapid initial growth, reaching its maximum value in a short distance, and then gradually decreases, as shown in Fig. 30. This is mainly influenced by the arc plasma pressure, as depicted in Fig. 31, where we extracted the maximum and minimum arc plasma pressure information along the symmetry axis of the arc plasma at 500ms for different keyhole depths. Fig. 31(a) shows the position information, Fig. 31(b) shows the arc plasma pressure values, and Fig. 31(c) shows the arc plasma pressure range encompassing maximum and minimum values. From Fig. 31(a), it can be observed that before and after considering the metal vapour, the maximum pressure points are the same, located at $z=7.1\text{ mm}$, near the tungsten electrode tip. However, the minimum pressure point increases with the keyhole

depth. As a result, the axial acceleration region of the arc plasma increases, and therefore, the axial velocity also increases with the keyhole depth.



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Fig. 31. Maximum and minimum pressure of arc along the symmetry axis.

As shown in Fig. 31(a), it is evident that the point of minimum pressure is not positioned at the bottom of the keyhole. As a result, between the point of minimum pressure and the keyhole bottom, the arc plasma pressure progressively rises, as depicted in Fig. 26, causing the arc plasma velocity to decelerate. This results in the axial velocity curve, which exhibits the characteristic pattern of first accelerating and then decelerating, as

depicted in Fig. 30. Further analysis of Fig. 31(a) reveals that, before and after considering the metal vapour, the positions of the maximum and minimum arc plasma pressure points are nearly identical. However, the values of the maximum and minimum arc plasma pressures differ, as depicted in Fig. 31(b). This leads to differences in the maximum and minimum arc plasma pressure differences, as depicted in Fig. 31(c). It can be seen that when metal vapour is not considered, the arc plasma pressure difference is larger, resulting in higher axial velocity, as shown in Fig. 30.

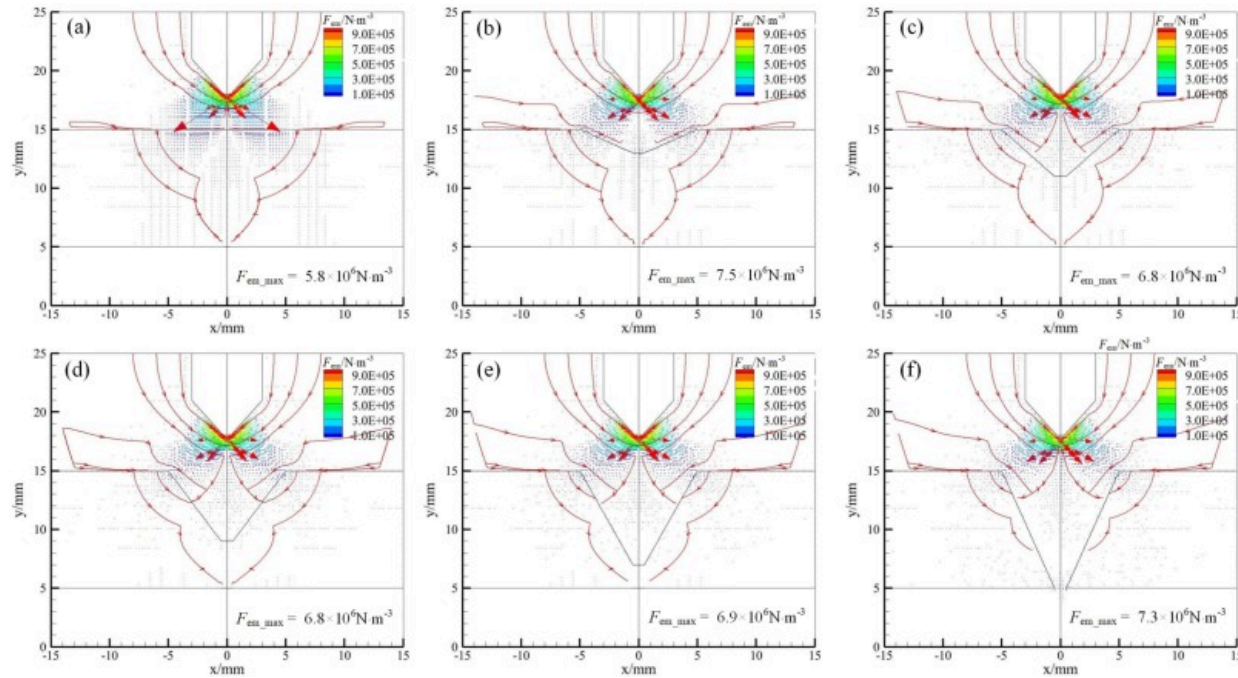
When the BM is not fully melted (i.e., the keyhole depth has not reached 10mm), the axial velocity of the arc plasma sharply slows down near the weld pool surface, generating exceptionally high stagnation pressure values. This is one of the primary factors that facilitate the creation of a keyhole within the weld pool in K-TIG welding. Once the BM is fully melted, the arc plasma axial velocity increases at the keyhole exit when metal vapour is not considered. This increase is due to the reduction in arc plasma pressure at this point, as depicted in the magnified view in Fig. 26(f), leading to the increase in axial velocity, and this is one of the key factors that enable the formation of a keyhole within the weld pool in K-TIG welding. When metal vapour is considered, the axial velocity increases significantly at keyhole exit. This is because the arc plasma pressure suddenly drops to $-15,511.95\text{Pa}$ at this point, as depicted in Fig. 26(f), resulting in a large arc plasma pressure difference. The arc plasma is thus rapidly accelerated, causing a sharp increase in axial velocity.

3.5. Effect of Fe metal vapour on electromagnetic force

As previously analyzed, after considering metal vapour, the arc plasma axial velocity decreases significantly. However, the factors influencing the axial velocity include not only the arc plasma pressure but also gravity and electromagnetic forces. Regardless of whether metal vapour is considered, the effect of gravity on the arc plasma remains the same, as it is a constant ($9.81\text{ m}\cdot\text{s}^{-2}$). Therefore, the variation in axial velocity must be caused by electromagnetic forces and arc plasma pressure. Since the arc plasma pressure

has already been discussed, we further extracted the spatial distribution map of the electromagnetic force on the arc plasma for further study.

As shown in [Fig. 32](#), the distribution of vectors and trajectory lines of the electromagnetic force acting on the arc plasma at different keyhole depths at 500ms, when metal vapour is not considered, are presented. The electromagnetic force in the tungsten electrode tip region is relatively large, much higher than in other regions. This is primarily because the current density at this point is high, as shown in [Fig. 20](#). According to Eqs. (2), (4), the magnitude of the electromagnetic force is closely related to the current density and magnetic flux density. Therefore, the electromagnetic force in this region is large, the largest value reach $7.5 \times 10^6 \text{N} \cdot \text{m}^{-3}$ when keyhole depth is 2mm. In other regions, as the arc plasma expands, the current density reduces, as depicted in [Fig. 20](#), leading to a corresponding reduction in electromagnetic force. In the BM domain, the current trajectory lines are diverging, as depicted in [Fig. 20](#), due to the minimal current density, the electromagnetic force vectors become barely discernible in the image.



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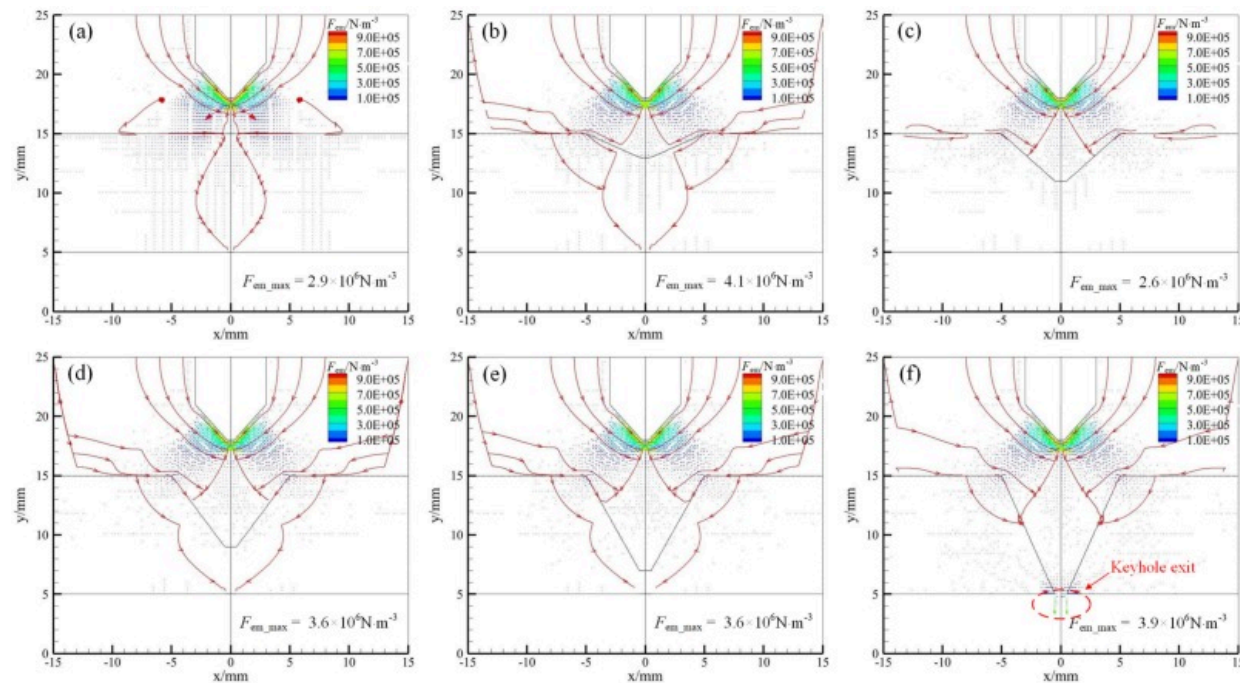
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Fig. 32. Electromagnetic force vector distribution (without metal vapour).

Furthermore, from the electromagnetic force trajectory lines in Fig. 32, it can be observed that the electromagnetic force primarily drives the arc plasma to move inward and downward. However, the electromagnetic force near the tungsten electrode is stronger, exerting a strong compressive effect on the arc plasma. In contrast, the electromagnetic force near the BM domain is weaker, resulting in a weaker compressive effect. As a result, the arc plasma profile presents a bell - shaped distribution.

As depicted in Fig. 33, when metal vapour is considered, the electromagnetic force vector distribution and trajectory lines of the arc plasma under varying keyhole depths at 500ms are illustrated. At this point, despite the maximum electromagnetic force still being situated at the tip of tungsten electrode, it is significantly reduced compared to when

metal vapour is not considered, with the maximum electromagnetic force approximately halved, only $4.1 \times 10^6 \text{ N} \cdot \text{m}^{-3}$ when keyhole depth is 2mm. This is mainly because, after considering metal vapour, the current density decreases, as depicted in Fig. 19, Fig. 21 and Fig. 24, leading to a reduction in electromagnetic force. The rest of the variation pattern follows the same trend as when metal vapour is not considered. However, when the keyhole depth is 10mm, the electromagnetic force at the keyhole exit is larger compared to when metal vapour is not considered, and its direction is vertically downward, as depicted in Fig. 33(f). Further arc plasma acceleration occurs due to the combined effect of keyhole exit electromagnetic force and arc plasma pressure, leading to a notable rise in axial velocity.

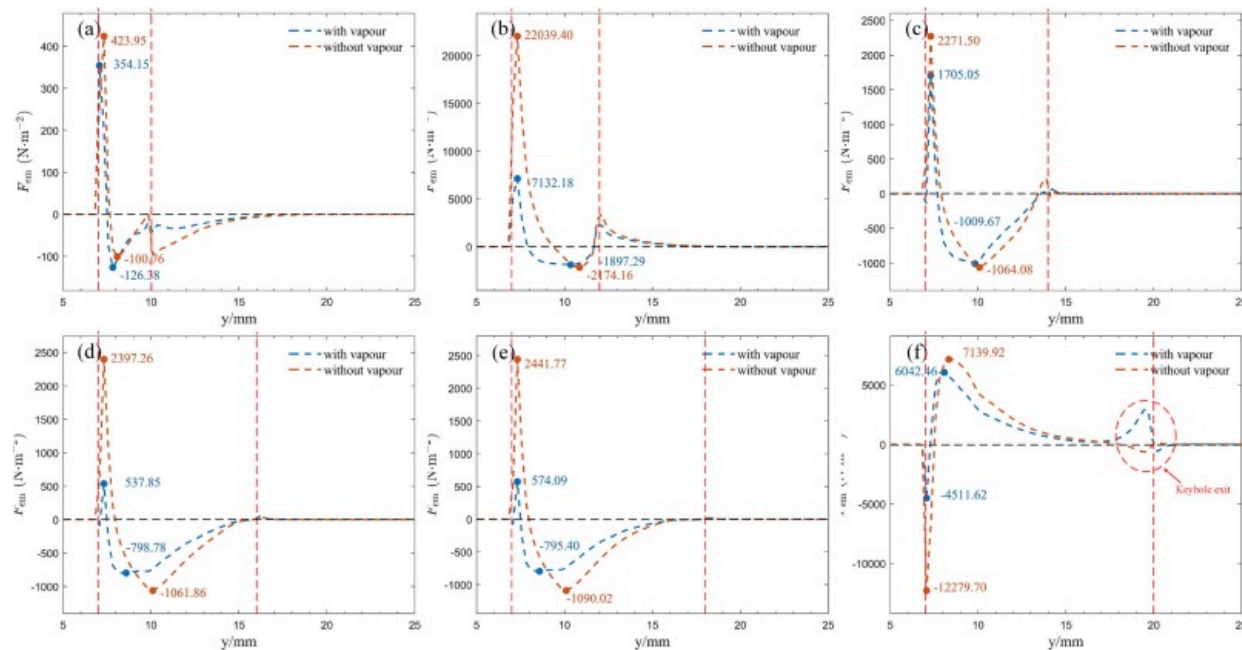


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Fig. 33. Electromagnetic force vector distribution (with metal vapour).

To further examine the distribution characteristics of the axial electromagnetic force within the arc plasma with and without considering metal vapour, we extracted the axial electromagnetic force curve along the axis at 500ms under different keyhole depths, as depicted in Fig. 34. From Fig. 34, it can be observed that the axial electromagnetic force near the tungsten electrode region is much larger than in other regions, and this force is positive, indicating that it accelerates the downward movement of the arc plasma. Subsequently, the electromagnetic force becomes negative and its magnitude decreases, gradually decelerating the arc plasma. This trend aligns perfectly with the change observed in the axial velocity distribution curve.



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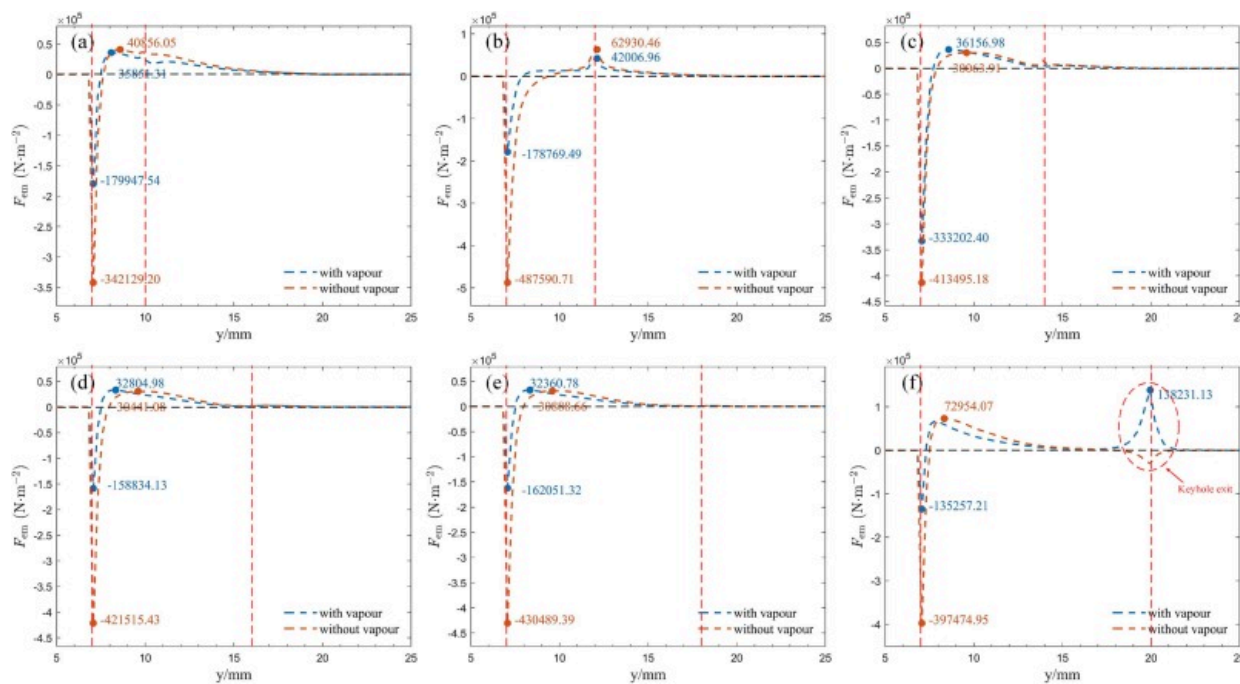
Fig. 34. Axial electromagnetic force distribution along the symmetry axis.

Furthermore, without considering metal vapour, the value of the axial electromagnetic force is larger, indicating a stronger accelerating effect, and thus the axial velocity is

higher. From Fig. 34, it can also be observed that when the keyhole depth is 0mm, the magnitude of the axial electromagnetic force is significantly lower than that observed at greater keyhole depths – measured at $423.95\text{N}\cdot\text{m}^{-3}$ and $354.15\text{N}\cdot\text{m}^{-3}$, respectively. Moreover, the difference in axial electromagnetic force before and after considering metal vapour is relatively small, only $69.8\text{N}\cdot\text{m}^{-3}$, as shown in Fig. 34(a). At this point, the difference in axial velocity is only $135.8\text{m}\cdot\text{s}^{-1}$, as depicted in Fig. 30(a). As the keyhole depth grows to 2mm, the axial electromagnetic force increases sharply, reached $22,039.4\text{N}\cdot\text{m}^{-3}$ and $7132.18\text{N}\cdot\text{m}^{-3}$ respectively, as depicted in Fig. 34(b), which elevates the axial velocity. At this point, the difference in axial electromagnetic force before and after considering metal vapour is relatively large, reach $14,907.22\text{N}\cdot\text{m}^{-3}$, resulting in a larger difference in axial velocity, which reaches $191.64\text{m}\cdot\text{s}^{-1}$, as depicted in Fig. 30(b). As the keyhole depth continues to increase, the difference in axial electromagnetic force remains significant, as depicted in Fig. 34(c)–(e), but smaller than keyhole depth is 2mm, and the difference in axial velocity is maintained around $200\text{m}\cdot\text{s}^{-1}$. When the keyhole depth reaches 10mm, the axial electromagnetic force initially takes a large negative value, $-12,279.70\text{N}\cdot\text{m}^{-3}$ and $-4511.62\text{N}\cdot\text{m}^{-3}$, respectively. At this point, the arc plasma should theoretically undergo reverse acceleration, but because the arc plasma pressure difference is positive, 1772.77Pa and 1366.62Pa respectively, as depicted in Fig. 26(f), the combined effects of both factors cause the axial velocity to continue increasing. With increasing distance from the tungsten electrode, the axial electromagnetic force becomes positive and, together with the arc plasma pressure difference, continues to accelerate the arc plasma. At this point, the axial electromagnetic force is much larger than at keyhole depths of 4, 6, and 8mm. Therefore, although the initial axial electromagnetic force is negative, the larger electromagnetic force results in the axial velocity quickly reaching almost the same value. Near the keyhole exit, as depicted in Fig. 34(f), the axial electromagnetic force becomes a large positive value when considering metal vapour, accelerating the arc plasma. In addition, the arc plasma pressure at the keyhole exit decreases, only $-15,511.95\text{Pa}$, as depicted in Fig. 26(f). Under the influence of the arc plasma pressure difference, the arc plasma also accelerates, causing a sharp increase in axial velocity, reach $1457.03\text{m}\cdot\text{s}^{-1}$. When metal vapour is not considered, the axial

electromagnetic force at the keyhole exit is negative, as depicted in Fig. 34(f), decelerating the arc plasma. However, due to the arc plasma pressure difference, the axial velocity still increases.

As depicted in Fig. 35, the radial electromagnetic force distribution curve along the axis at the 500ms moment for different keyhole depths, both with and without considering metal vapour, is presented. From Fig. 35, it can be seen that near the tungsten electrode region, the radial electromagnetic force is much larger than in other regions, reaching $-342,129.20\text{N}\cdot\text{m}^{-3}$ and $-179,947.54\text{N}\cdot\text{m}^{-3}$ respectively when the keyhole depth is 0mm, and its direction is negative, indicating a strong compressive effect on the arc plasma.



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Fig. 35. Radial electromagnetic force distribution along the symmetry axis.

As the position moves farther from the tungsten electrode, the radial electromagnetic force gradually decreases, and its direction shifts to positive. When the keyhole depth is 0mm, the maximum values reach $40,856.05 \text{ N}\cdot\text{m}^{-3}$ and $35,851.34 \text{ N}\cdot\text{m}^{-3}$, respectively. This indicates that the radial electromagnetic force at this stage acts to drive the arc plasma outward, promoting its expansion. Therefore, the arc plasma has a bell - shaped distribution, exhibiting compressive behavior in the upper zone near the tungsten electrode and expansive characteristics in the lower zone adjacent to the BM.

Additionally, it can be observed that after incorporating the effects of metal vapour, the radial electromagnetic force is significantly reduced. For instance, at a keyhole depth of 4mm, it decreases from $-413,495.18 \text{ N}\cdot\text{m}^{-3}$ to $-333,202.40 \text{ N}\cdot\text{m}^{-3}$ – approximately a 25% reduction, as shown in [Fig. 35\(c\)](#). At other keyhole depths, the reduction is even more substantial, averaging around 75%. For example, at a depth of 2mm, the force drops from $-487,590.71 \text{ N}\cdot\text{m}^{-3}$ to $-178,769.49 \text{ N}\cdot\text{m}^{-3}$, as shown in [Fig. 35\(b\)](#). This weakening of the compressive force on the arc plasma results in a broader spread of the isothermal contours in the corresponding region. When approaching the BM side, after considering metal vapour, the radial electromagnetic force expanding the arc plasma is even lower, causing the isothermal lines to contract, as depicted in [Fig. 15](#). When keyhole depth is 10mm, in the keyhole exit region, the radial electromagnetic force without considering metal vapour is negative, leading to compression of the arc plasma. However, after considering metal vapour, the radial electromagnetic force suddenly increases and becomes positive, resulting in the expansion of the arc plasma. As a result, a significant difference in the isothermal lines appears, as depicted in [Fig. 15\(f\)](#).

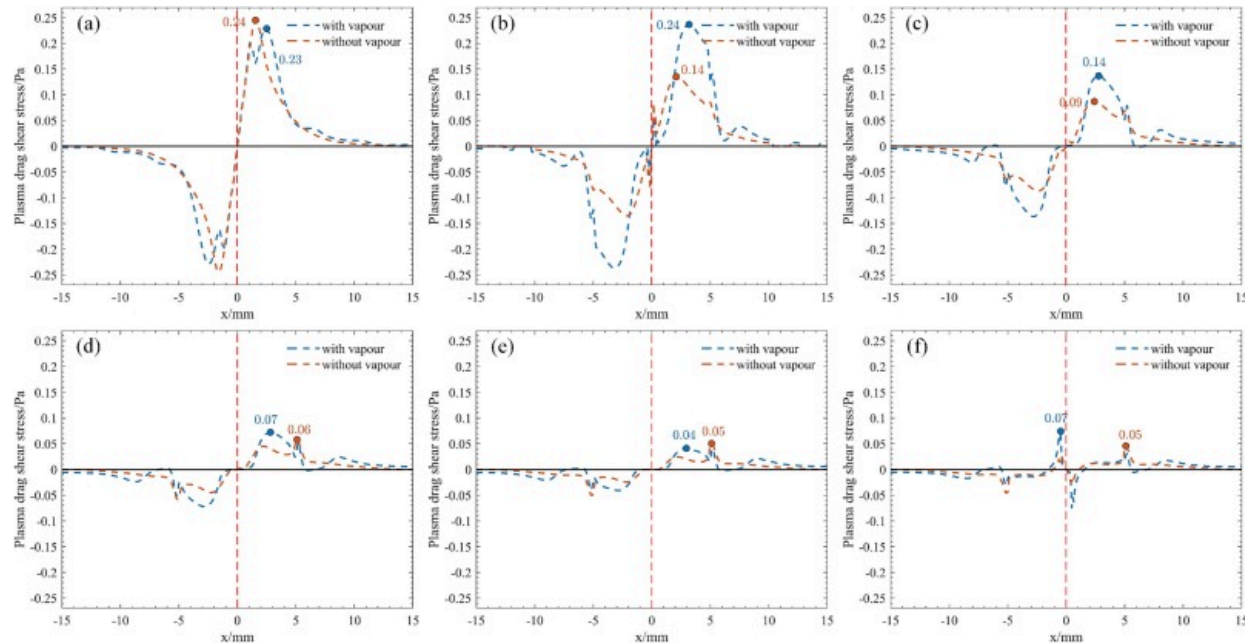
3.6. Effect of Fe metal vapour on weld pool surface shear stress

According to our previous study [[42](#)], the Marangoni shear stress – arising from surface tension gradients induced by temperature variations along the weld pool surface – can be estimated based on thermophysical principles. This shear stress is directly proportional to the surface tension gradient, which is itself a function of the temperature gradient.

To assess this effect, a post - processing analysis was conducted using Tecplot to extract the temperature gradient distribution along the weld pool surface from the simulated temperature field. A constant surface tension temperature coefficient ($\partial\gamma/\partial T$) of $-0.0004\text{N}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, corresponding to the Q345 BM, was adopted based on material property data in Ref. [42]. By combining the calculated temperature gradients with this coefficient, the distributions of both Marangoni shear stress and plasma drag shear stress on the weld pool surface were derived.

It should be noted that this modeling strategy neglects the dynamic deformation of the weld pool surface, which may compromise the accuracy of the predicted weld pool morphology under the influence of metal vapour. Such an assumption may mask the actual degree to which vapour - induced modifications in arc plasma characteristics affect the evolution of the weld pool. In particular, it omits the transient surface response driven by plasma drag and Marangoni shear stresses, therefore, it is impossible to obtain accurate Marangoni and plasma drag flow shear stresses. However, by assuming a fixed surface, it is still possible to obtain a qualitative assessment of the surface - driven flow mechanisms, such as Marangoni convection. Similar simplifications have been successfully adopted in previous studies [25, 42, 75], which demonstrated reasonable agreement with experimental trends. Therefore, although limited in scope, this method remains valuable for isolating and analyzing arc - induced thermocapillary effects.

As shown in Fig. 36, the distribution of plasma drag shear stress along the keyhole surface at 500ms is presented for various keyhole depths, with and without the inclusion of metal vapour. The overall magnitude of the plasma drag shear stress is relatively low, reaching only approximately 0.2Pa. This is primarily because plasma drag shear stress is governed by both the velocity gradient and the viscosity coefficient. In the present model, which neglects deformation of the BM, the velocity gradient along the keyhole surface remains small. Furthermore, the high arc plasma temperature in this region lowers the viscosity coefficient, thereby further reducing the plasma drag shear stress.



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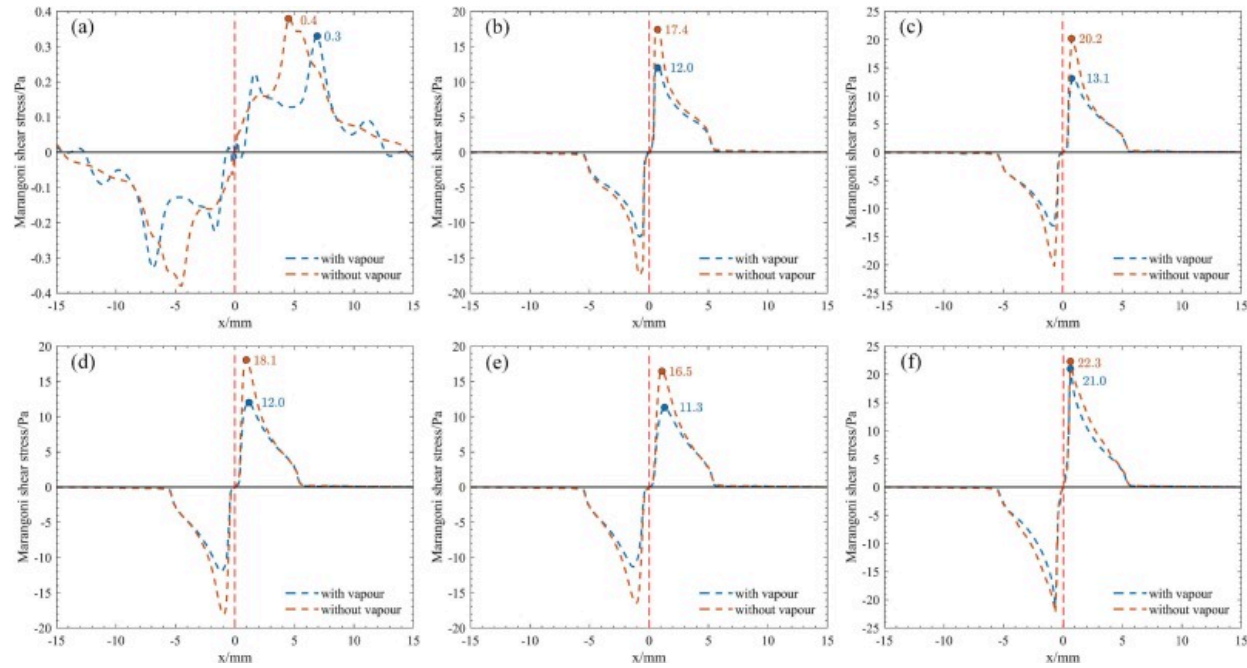
Fig. 36. Distribution of plasma drag shear stress on the keyhole surface.

Additionally, Fig. 36(a)–(c) reveals that when the keyhole depth is relatively shallow, the difference in plasma drag force between cases with and without metal vapour becomes more pronounced. The absolute magnitude of the force is also higher compared to deeper keyhole cases [Fig. 36(d)–(f)]. This observation aligns with Fig. 41, which shows that metal vapour is visibly distributed near the upper surface when the keyhole is shallow, but becomes negligible at greater depths. These findings suggest that metal vapour exerts a significant influence on plasma drag shear stress. Further support for this conclusion is provided by Fig. 5(c), which indicates that the addition of metal vapour substantially alters the viscosity of the arc plasma, thereby modifying the distribution of plasma drag shear stress.

[Fig. 36](#) further demonstrates that plasma drag shear stress exhibits a decreasing trend with increasing keyhole depth. When the BM is close to complete penetrating, the plasma drag shear stress is reduced to approximately one - third of its initial value, about 0.7Pa. Additionally, when the keyhole depth is small, as depicted in [Fig. 36\(a\)–\(c\)](#), the plasma drag shear stress mainly drives the weld pool metal to flow to the sides. On the left side of the symmetry axis, the plasma drag shear stress is negative, driving the weld pool metal to move in the negative direction. Conversely, at the right - hand side relative to the axis of symmetry, the plasma drag shear stress is positive, leading the weld pool metal to flow in the positive direction.

Moreover, as the radial distance increases, the plasma drag shear stress initially increases and then decreases. This trend suggests that, when the keyhole is shallow, the plasma drag shear stress near the center and close to the keyhole region increases rapidly, promoting the outward flow of molten metal within the weld pool. However, as the keyhole deepens, the molten metal at the bottom must overcome gravitational forces to move upward before it can flow outward along the keyhole walls. This additional resistance slows down the flow velocity, leading to reduced plasma drag shear stress, which also exhibits irregular fluctuations, as shown in [Fig. 36\(d\)–\(f\)](#).

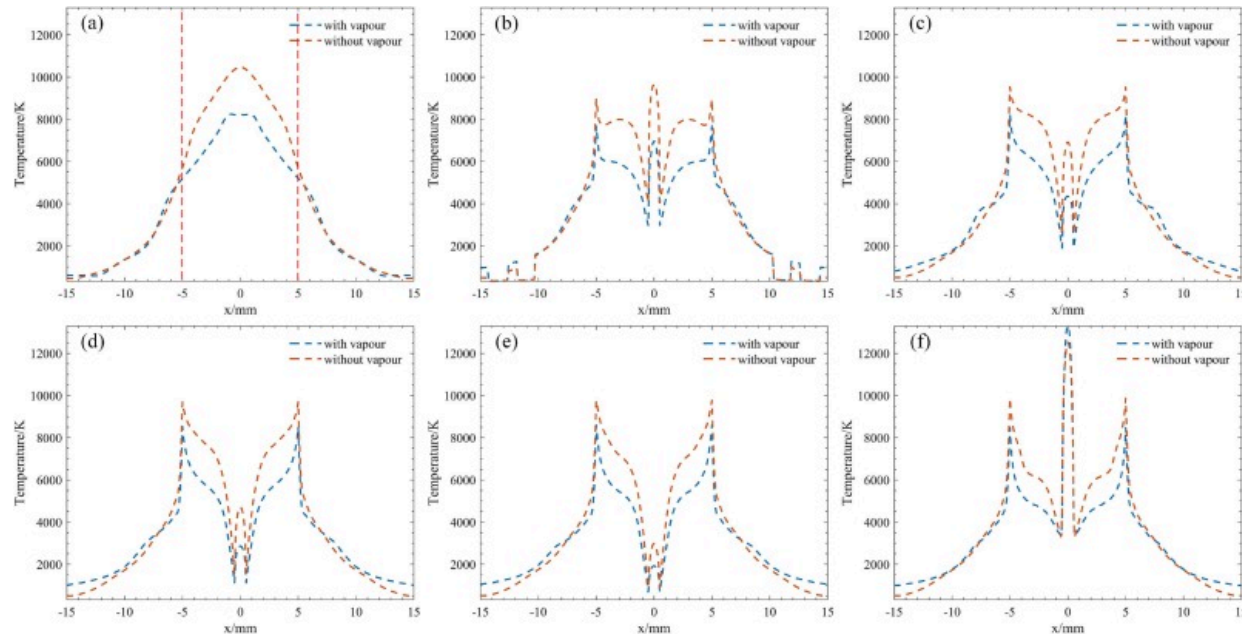
As depicted in [Fig. 37](#), the curve of Marangoni shear stress distribution on the keyhole surface at the 500ms moment for different keyhole depths, both with and without considering metal vapour, is presented. According to our previous research [\[42\]](#), Marangoni shear stress is linked to temperature gradients, and the sharp temperature change from the arc's center to its outer regions, as shown in [Fig. 38](#), leads to a high Marangoni shear stress value.



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Fig. 37. Distribution of Marangoni shear stress on the keyhole surface.



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Fig. 38. Distribution of temperature profile on the keyhole surface.

When the keyhole depth is 0mm, the surface of the BM is flat, and the temperature variation of the arc plasma is relatively slow, as depicted in Fig. 38(a). Therefore, the temperature gradient is small, and the Marangoni shear stress is correspondingly small, only 0.4 and 0.3Pa respectively, as seen in Fig. 37(a). After the keyhole forms, a large amount of arc plasma enters the keyhole, and upon reaching the bottom of the keyhole, it changes direction and flows outward along the keyhole wall. As a result, the heat transfer on the keyhole wall changes, leading to a sharp temperature variation, as depicted in Fig. 38(b)–(f). This results in a large temperature gradient and, consequently, a larger Marangoni shear stress, reached above 10Pa, as depicted in Fig. 37(b)–(f). Moreover, the inclusion of metal vapour results in a reduction in Marangoni shear stress. This can likely be attributed to the enhanced electrical conductivity of the arc plasma caused by the presence of metal vapour, which in turn reduces Joule heating. As a consequence, the

temperature gradient across the weld pool surface decreases, leading to a diminished Marangoni shear stress.

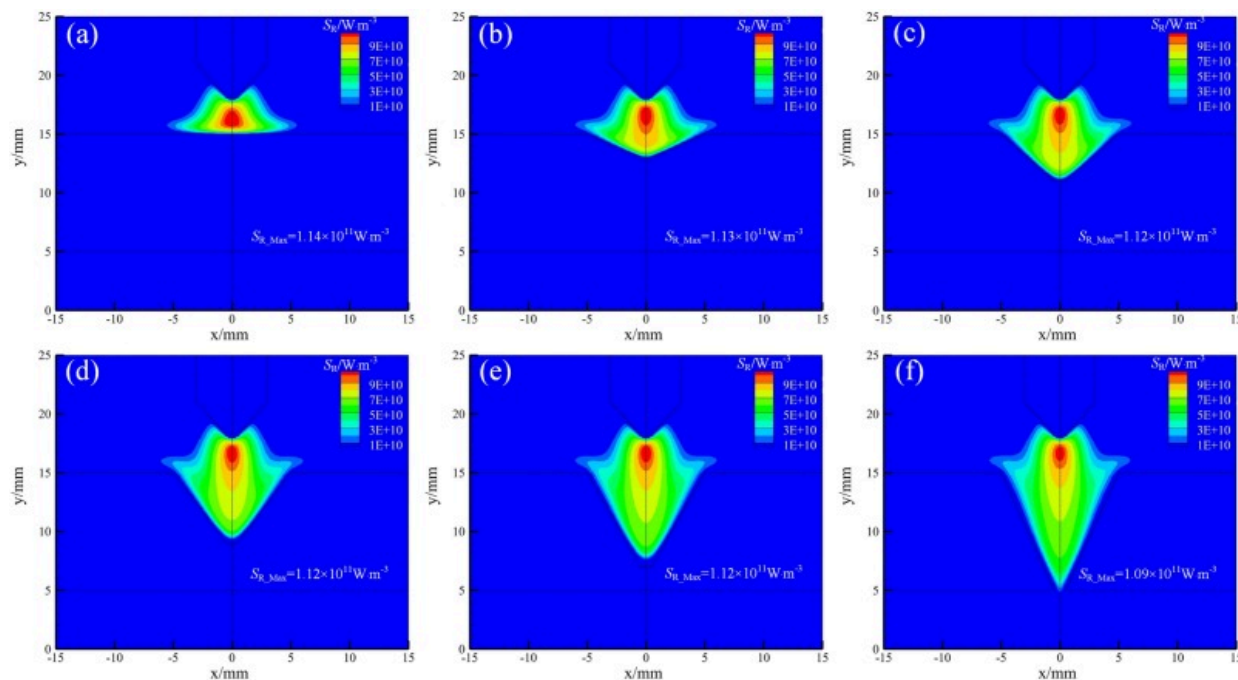
In the central region of the keyhole, the temperature variation is most pronounced, and as a result, the Marangoni shear stress increases rapidly to its maximum value. As the force moves outward, the temperature variation gradually decreases, and thus the Marangoni shear stress also decreases gradually.

Furthermore, the Marangoni shear stress primarily drives the weld pool metal to flow to both sides, exhibiting different directional effects on either side of the central symmetry axis. When the keyhole depth is 0mm, the Marangoni shear stress is small; however, once the keyhole forms, the Marangoni shear stress increases rapidly. This phenomenon is primarily attributed to the formation of the keyhole, which, according to the principle of minimum voltage, causes the current path to concentrate near the upper edge of the keyhole entrance, as shown in [Fig. 21](#) and [Fig. 22](#). This concentration leads to significant Joule heating in that region. In contrast, the interior of the keyhole relies mainly on thermal conduction from the arc plasma entering the cavity, resulting in comparatively lower heat input. Consequently, the temperature distribution within the keyhole, as illustrated in [Fig. 38](#), exhibits a pronounced temperature gradient at the center of the keyhole. This gives rise to the characteristic Marangoni shear stress distribution shown in [Fig. 37](#), which drives the molten metal in the weld pool to flow laterally outward, thereby promoting keyhole formation — a process that plays a critical role in the realization of stable K-TIG welding.

3.7. Effect of Fe metal vapour on radiation loss

As depicted in [Fig. 39](#), the radiation loss distribution cloud map of the arc plasma under varying keyhole depths at 500ms, without considering the impact of metal vapour, is presented. [Fig. 5\(f\)](#) demonstrates a positive correlation between arc plasma temperature and arc plasma radiation loss. Therefore, the radiation loss is higher in the high - temperature core region of the arc plasma and lower in the peripheral low - temperature

regions, with the maximum radiation loss reaching approximately $1.1 \times 10^{11} \text{ W} \cdot \text{m}^{-3}$. From Fig. 39, it can also be observed that the maximum radiation loss of the arc plasma remains almost unchanged with varying keyhole depths. This is primarily because the maximum temperature remains similar, and it is concentrated near the tungsten electrode region, as depicted in Fig. 9. Fig. 39 reveals that in the absence of keyhole (0mm), the high radiation loss region is relatively wide. As the keyhole forms, the high radiation loss region becomes narrow and elongated. After the keyhole is formed, the arc plasma moves into it and undergoes compression, which also compresses and elongates its high - temperature region, thereby altering the radiation loss.



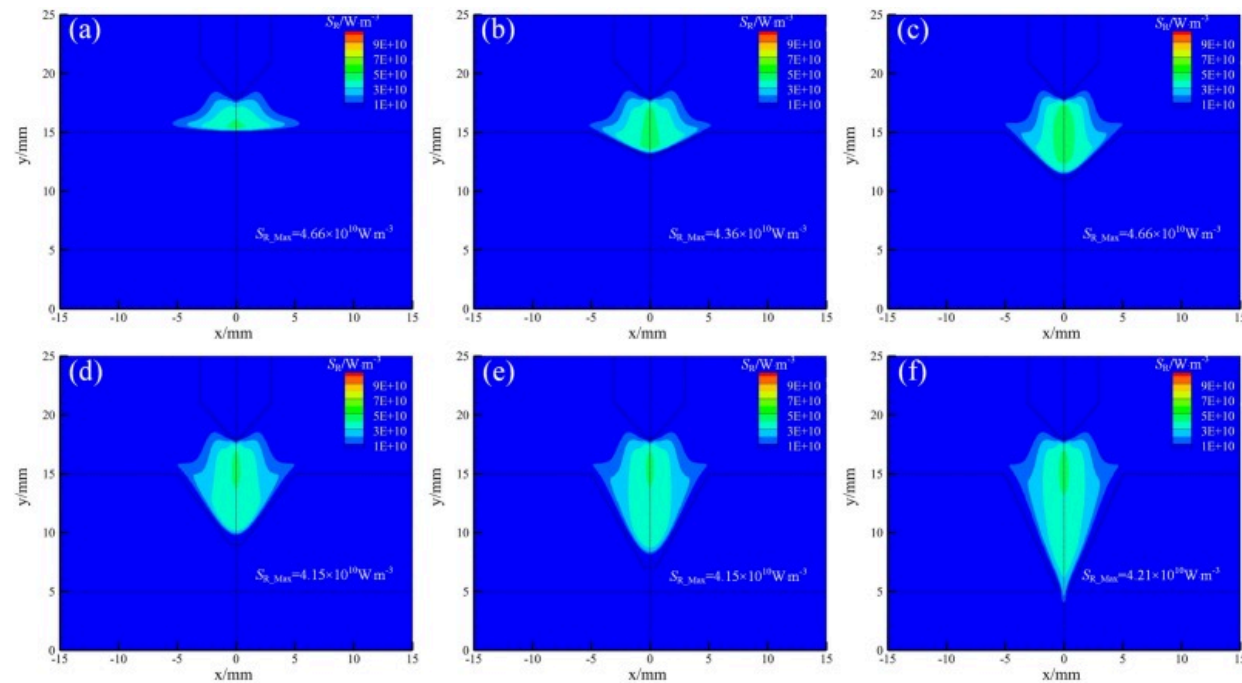
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Fig. 39. Distribution of radiation loss on the keyhole surface (without metal vapour).

As shown in Fig. 40, the distribution of radiation loss for the arc plasma at 500ms, considering the effects of metal vapour, is presented for varying keyhole depths. The

presence of metal vapour leads to an increase in the medium - temperature range (5000–15,000K) electrical conductivity of the arc plasma, leading to a reduction in Joule heating, i.e., a decrease in temperature. Consequently, the radiation loss is also significantly reduced, with a maximum value of $4.5 \times 10^{10} \text{ W} \cdot \text{m}^{-3}$, which is only about 40% of that without metal vapour. Meanwhile, similar to the case without metal vapour, since the maximum temperature change is relatively small, the maximum radiation loss remains nearly constant with varying keyhole depths.



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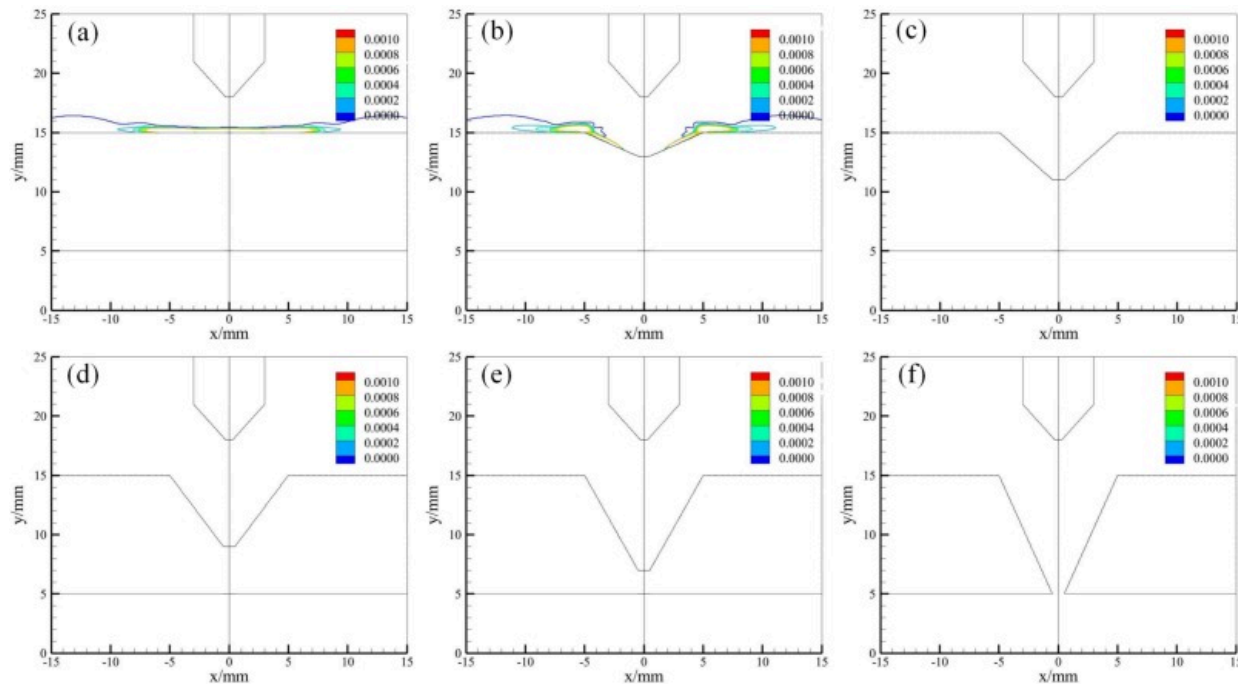
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Fig. 40. Distribution of radiation loss on the keyhole surface (with metal vapour).

3.8. Effect of Fe metal vapour distribution

As depicted in Fig. 41, the distribution of metal vapour on the BM anode surface at 500ms for varying keyhole depths is displayed. It can be observed that when the keyhole depth is

0mm or 2mm, metal vapour is clearly visible on the anode surface. However, as the keyhole depth increases further, no metal vapour is observed under the same color scale, indicating that the surface temperature is too low to generate metal vapour or that the vapour concentration is minimal. Fig. 7 and Fig. 8 also reveal that with the keyhole depth increasing, the heat transferred to the BM decreases, which explains the absence of metal vapour distribution.



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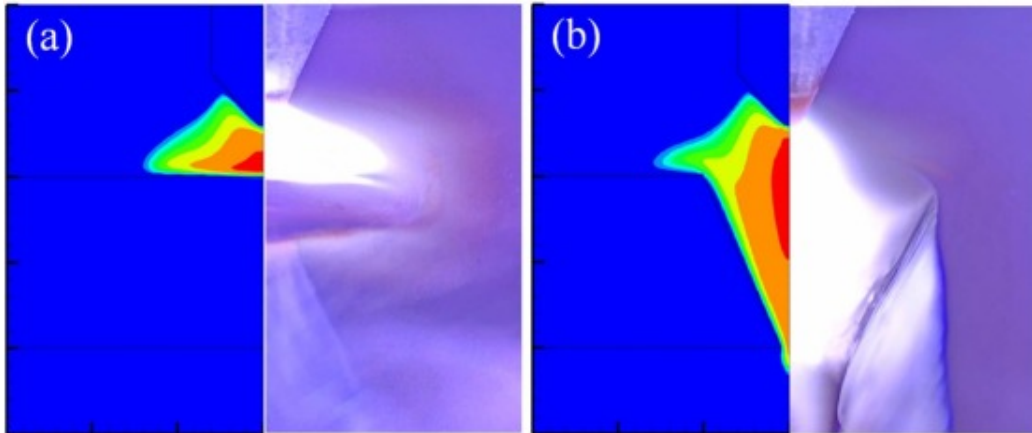
Fig. 41. Distribution of metal vapour under different keyhole depth.

Further analysis shows that the metal vapour expands radially on the BM anode top surface. This is primarily due to the arc plasma moving from the tungsten electrode tip toward the anode. Upon reaching the anode, it expands radially because of the obstruction of the BM, causing the metal vapour to spread horizontally along the surface. Additionally, when the keyhole depth attains 2mm the metal vapour is primarily

concentrated at the upper corner of the keyhole, with no metal vapour observed at the keyhole bottom. Previous studies have shown that current trajectories are mainly distributed around the keyhole corners, with a sparse distribution at the bottom, leading to lower heat generation. Since metal vapour formation is closely related to temperature, it predominantly appears in regions of high current density. As the keyhole depth increases further, Fig. 22(c)–(f) shows that, although the peak current density remains high, its distribution range becomes narrower. Consequently, the conditions for metal vapour generation are not met, or the generated amount is too low to be observed, explaining the absence of visible metal vapour distribution.

3.9. Arc plasma profile verification

To verify the model's accuracy, HDR camera images of the arc plasma profile were recorded for keyhole depths of 0mm and 10mm. These experimental results were contrasted with the simulation outcomes, as illustrated in Fig. 42. Due to the difficulty in precisely matching arc plasma images with actual keyhole depths for other conditions, only these two representative cases were selected for comparison. At without keyhole (0mm), the BM blocks the arc plasma, forcing it to spread outward when it reaches the surface. This results in a wider arc plasma shape, as depicted in Fig. 42(a). When the BM is fully penetrated and a stable keyhole structure is established, mechanical compression by the keyhole confines the arc plasma, leading to radial contraction and axial elongation of the arc plasma column, as depicted in Fig. 42(b). The consistency between the simulated and experimentally observed arc plasma profiles confirms the accuracy of the proposed model.



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Fig. 42. Comparison of the K-TIG welding arc profiles under different keyhole depth: (a) 0mm (b) 10mm.

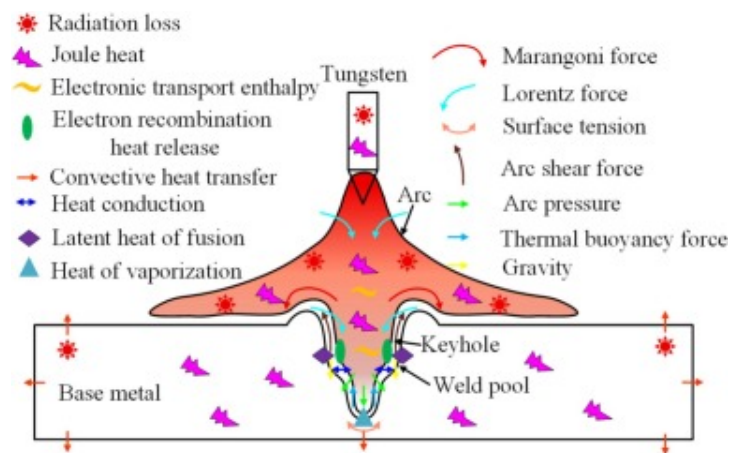
3.10. Experimental limitations and directions for future validation

Due to inherent experimental constraints, it is currently not feasible to directly measure or isolate the influence of metal vapour on weld pool depth and morphology. The generation of metal vapour during welding is intrinsically uncontrollable, as it occurs spontaneously once the local surface temperature exceeds the vaporization point of constituent elements. Consequently, establishing a “vapour - free” baseline for direct experimental comparison remains impractical.

Moreover, due to the current lack of advanced diagnostic tools — such as high - speed optical emission spectroscopy or two - color pyrometry — the arc plasma core temperature and the local Fe vapour concentration cannot yet be experimentally validated. As a result, the consistency between the simulated arc plasma temperatures (considering vapour effects) and the actual physical conditions remains to be fully established.

Additionally, the present model assumes a static weld pool surface and neglects weld pool deformation, which may limit the physical accuracy of the predicted weld pool shape under varying metal vapour conditions. This simplification could obscure the true extent to which vapour - induced changes in arc plasma behavior influence weld pool evolution. In particular, the dynamic response of the weld pool to arc plasma forces — such as arc plasma pressure, plasma drag shear stress, and Marangoni shear stress — is not captured.

Based on the above analysis, it can be inferred that under the action of these forces, the molten metal within the keyhole tends to flow outward and upward. As the displaced metal accumulates near the top surface, it leads to the gradual formation of a curved keyhole wall, thereby transforming the arc plasma - keyhole interface from an idealized planar slope into a curved boundary, as illustrated in Fig. 43. This morphological change can significantly affect the local arc plasma - keyhole interaction, including heat transfer, flow resistance, and the distributions of plasma drag and Marangoni shear stresses. Therefore, while the static surface assumption provides a simplified framework for isolating key physical mechanisms, it inevitably introduces deviations from the actual dynamic behavior of the weld pool.



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Fig. 43. Schematic diagram of arc plasma - keyhole - weld pool interaction.

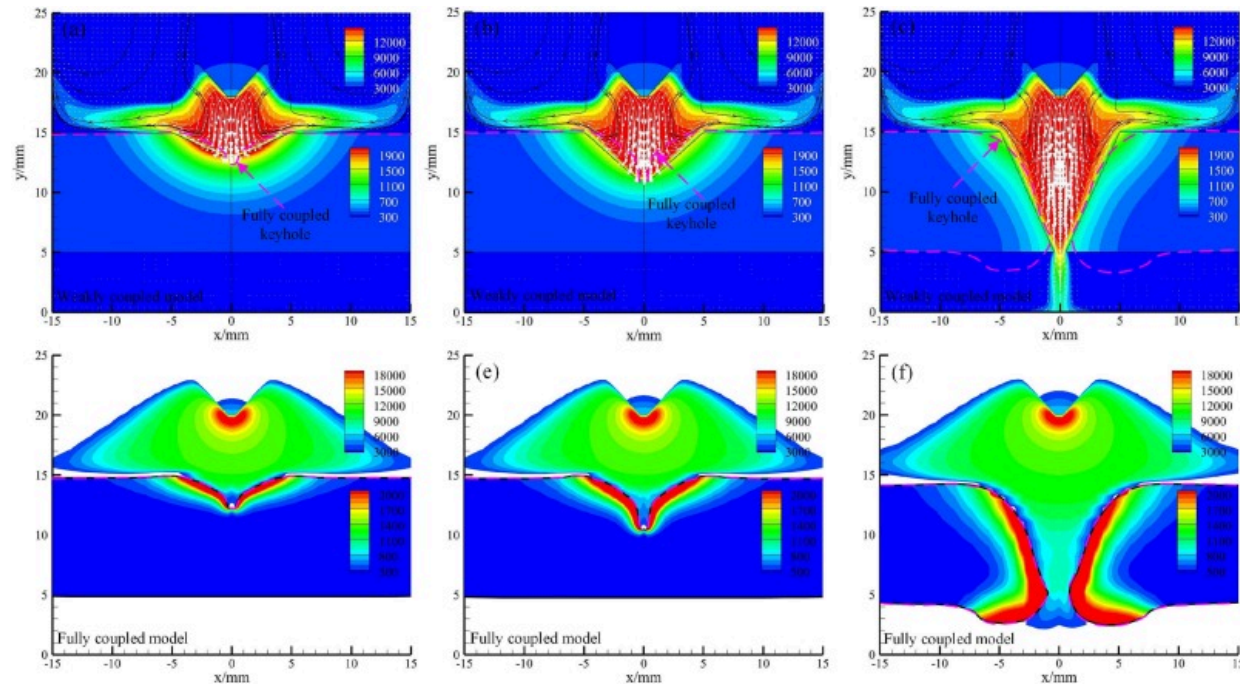
As the keyhole depth evolves, this curved interface alters the interaction between the arc plasma and the keyhole wall, thereby influencing heat transfer mechanisms and modifying both the plasma drag shear stress and Marangoni shear stress. Specifically, under the current assumption of a planar keyhole wall, the arc plasma can flow outward along the slope with minimal resistance. However, in a more realistic scenario where the wall is curved, the arc plasma flow experiences varying degrees of obstruction – accelerating over shallow regions and decelerating in steeper sections. This differential flow behavior alters the local heating time, leading to spatial variations in the temperature gradient along the keyhole surface, which in turn affects the magnitude and distribution of Marangoni stresses.

Moreover, the presence of a curved interface can significantly hinder arc plasma flow, thereby modifying the plasma drag shear stress distribution. The arc plasma pressure distribution also shifts with changes in keyhole geometry, leading to feedback effects that further deform the keyhole shape. Once the keyhole is formed, the molten metal flows outward and accumulates near the outlet at the upper surface of the BM, creating a convex bulge, as shown in [Fig. 43](#). This region becomes a convergence point for arc plasma current trajectories, resulting in localized enhancement of Joule heating. In addition to promoting metal vaporization, the altered thermal field at this convex region further influences the distribution of both Marangoni and plasma drag shear stresses. Since the current model does not account for this deformation, deviations between the simulated force distributions and their actual counterparts may arise near the keyhole exit.

To further substantiate the assumptions adopted in this study, a fully coupled arc plasma - weld pool model that incorporates dynamic surface deformation and molten metal flow has been developed. This model captures the transient evolution of the keyhole and enables a more comprehensive representation of the arc plasma - weld pool interaction. As illustrated in [Fig. 44](#), the keyhole profile predicted by the fully coupled model closely

matches the profile obtained under the assumption of a static weld pool surface.

Although a detailed analysis of the fully coupled model is beyond the scope of the present work, this result provides strong supplementary evidence supporting the validity of the simplified assumptions employed herein. Further elaboration and validation of this model will be presented in our future studies.



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Fig. 44. Comparison of keyholes profile (a–c) weakly coupled (d–f) fully coupled.

Furthermore, future work will aim to integrate in - situ diagnostic techniques and apply design-of - experiment (DOE) approaches – such as response surface methodology (RSM) [76] – to systematically evaluate the sensitivity of key parameters, including metal vapour concentration, on arc plasma characteristics and weld penetration. These efforts will contribute to more rigorous experimental validation and further enhance the predictive capability of the simulation framework.

4. Conclusion

In this study, a 2D unified tungsten electrode - arc plasma - keyhole coupled multiphysics simulation model was developed, in which the BM surface was assumed to remain undeformed and keyholes of varying preset depths were adopted to accelerate computation, for investigating the influence of Fe metal vapour on arc plasma - keyhole transport mechanisms in K-TIG welding. The key conclusions are as follows:

1. The presence of metal vapour was found to increase electrical conductivity in the medium -temperature range (5000–15,000K), thereby altering the current density distribution, reducing Joule heating and radiation losses, and lowering the maximum arc plasma temperature from 19,500K to 14,500K. Consequently, low - temperature isotherms near the electrode expanded outward, whereas high-temperature isotherms adjacent to the BM contracted inward.
2. Metal vapour generation is localized and depth - dependent, occurring predominantly near the upper corner of the keyhole where the current density is highest in BM domain, generates a large amount of Joule heat, causing the BM to vaporization and evaporate. The resulting increase in electrical conductivity leads to a redistribution of the current density, which in turn reduces both the axial and radial components of the electromagnetic force, decreases the arc plasma pressure from 1700Pa to 1400Pa, and lowers the axial plasma velocity by approximately $200\text{m}\cdot\text{s}^{-1}$ following keyhole formation. Furthermore, the presence of metal vapour modifies plasma drag and alters the distribution of Marangoni shear stress, thereby influencing weld pool flow patterns and reshaping the keyhole morphology.
3. The undeformed BM assumption with preset keyholes was validated using a fully coupled arc plasma – keyhole - weld pool model. The simulated keyhole geometries exhibited strong agreement with the preset profiles, and the simulated arc plasma profiles at representative keyhole depths

(0mm and 10mm) closely matched the experimental observations, confirming the accuracy and reliability of the simulation approach.

CRediT authorship contribution statement

Jiatong Zhan: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Yonghua Shi:** Writing – review & editing, Writing – original draft, Funding acquisition. **Xiongyue Ye:** Writing – review & editing, Software, Methodology, Investigation. **Shaotao Zhong:** Writing – review & editing, Validation, Methodology. **Zhizhong Liu:** Software, Methodology, Investigation. **Rong Zhang:** Writing – review & editing, Validation. **Yuchao Zhou:** Writing – review & editing, Methodology. **Zuxin Liang:** Writing – review & editing, Methodology, Investigation.

Consent to participate

Not applicable.

Consent for publication

All authors have read and agreed to the published version of the manuscript.

Ethics approval

The content studied in this article belongs to the field of metal processing and does not involve humans and animals. This article strictly follows the accepted principles of ethical and professional conduct.

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Declaration of competing interest

The authors declare no competing interests.

Data availability

All data collected and analyzed during this study are presented in this manuscript.

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